

Bones of the Lower Limb

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General advise for studying any bone

If it is long bone: study it by order; upper end, shaft and lower end. From all aspects ant, posterior, medial and lateral also know its borders.

If it is flat bone (e.g ribs) study its anterior end, posterior ends and shafts. Also borders if have? Upper & lower borders / surfaces--- inner and outer.

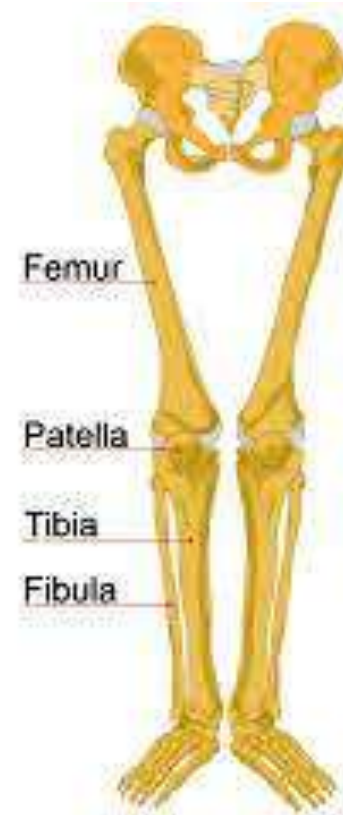
Function of LLs

The primary function of the lower limbs **is to support the weight of the body and to provide a stable foundation in standing, walking, and running; they have become specialized for locomotion.**

Organization of the lower limb

LLs are divided into the :

- ☐ **Gluteal region (buttock)**
- ☐ **Thigh**
- ☐ **Knee**
- ☐ **Leg**
- ☐ **Ankle**
- ☐ **Foot**



Bones of the gluteal region (Hip bone)

1-Hip bone:

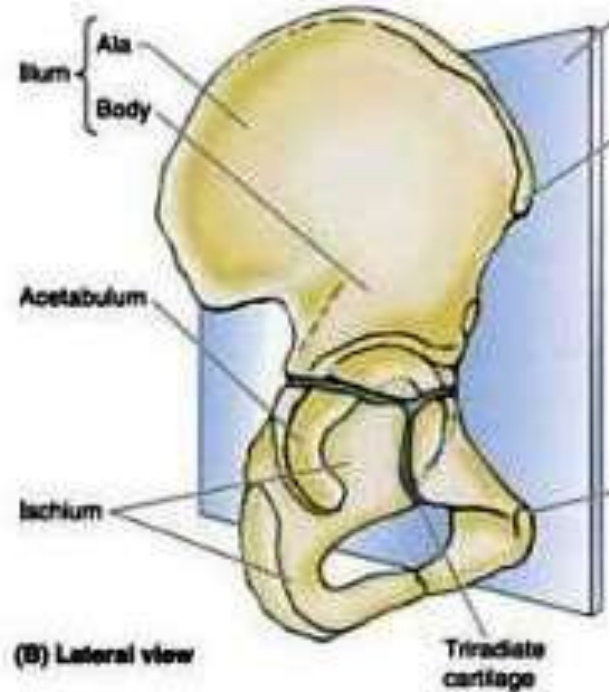
Composed of (3) bones
meet each other at

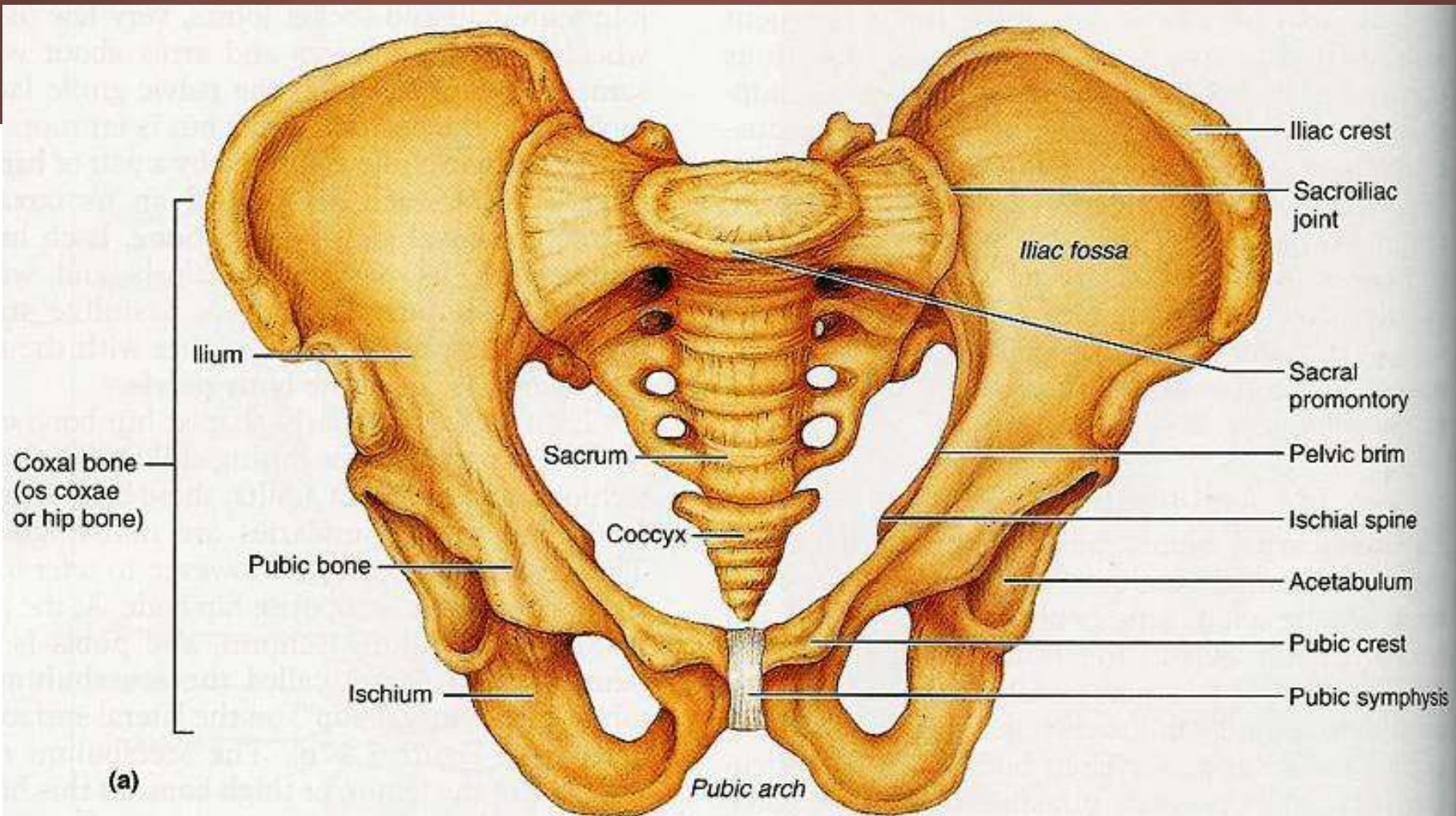
acetabulum:

1- ilium

2- Ischium

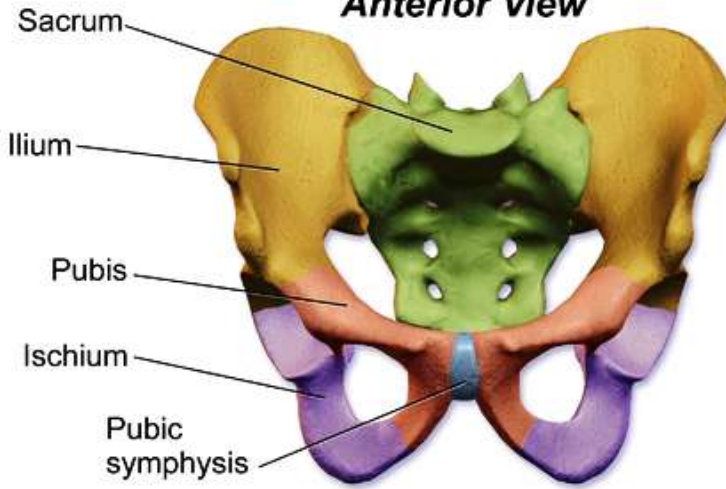
3- Pubis



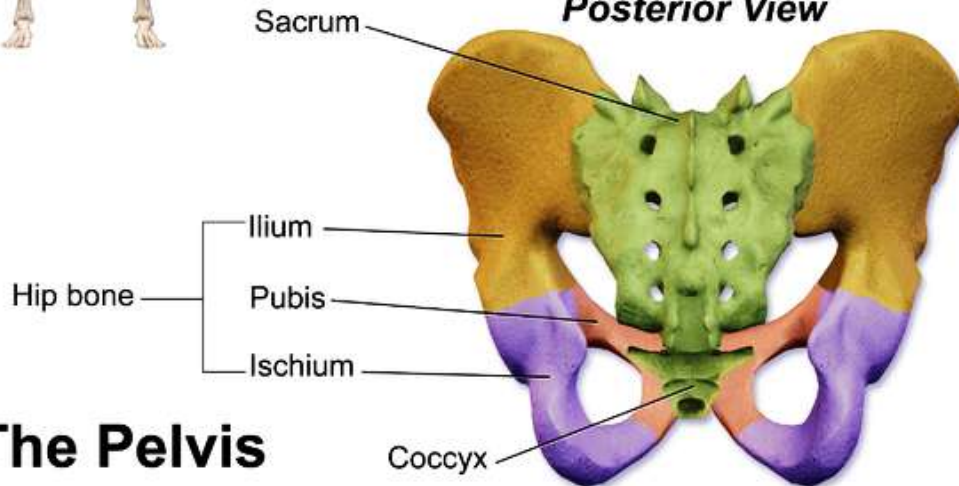




Anterior View



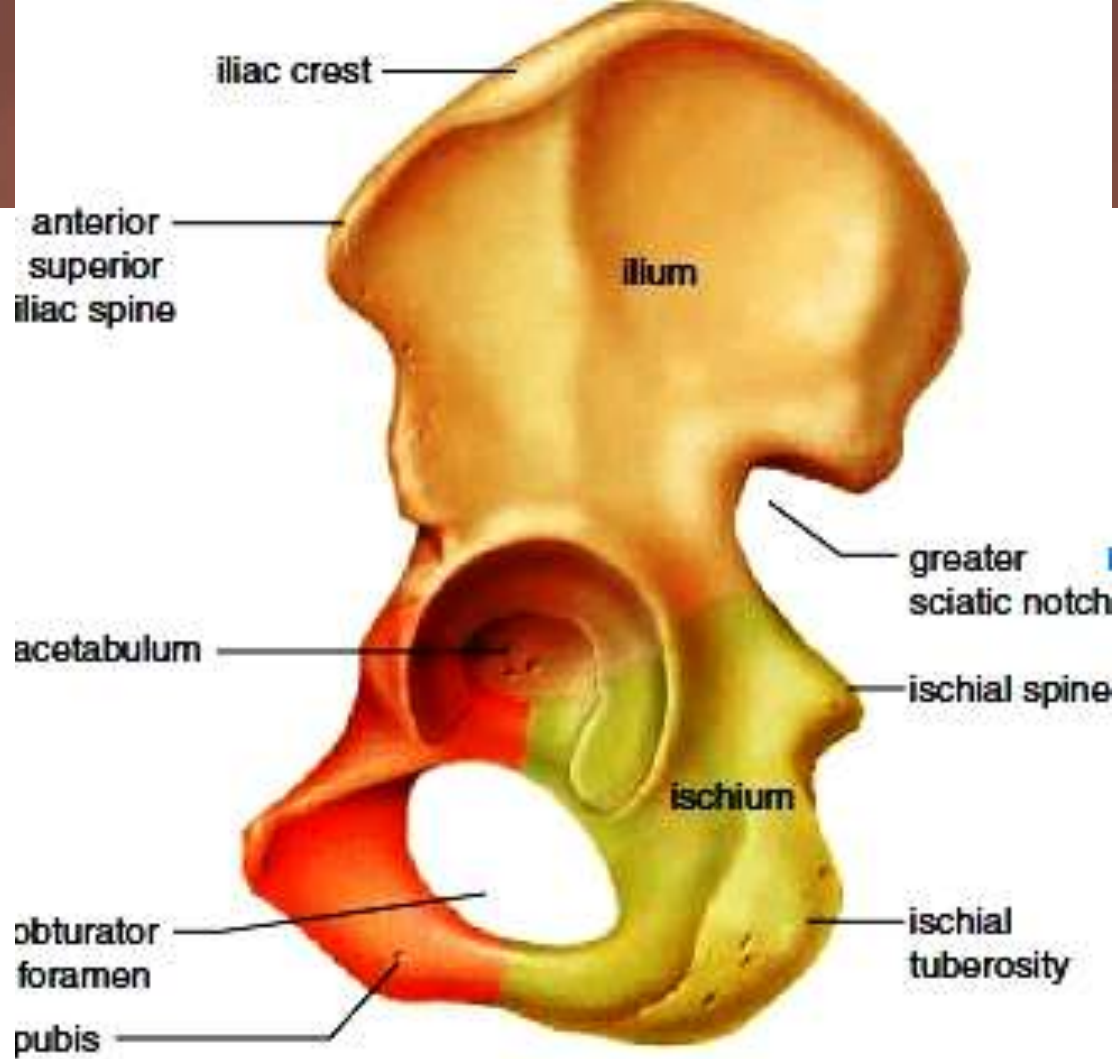
Posterior View



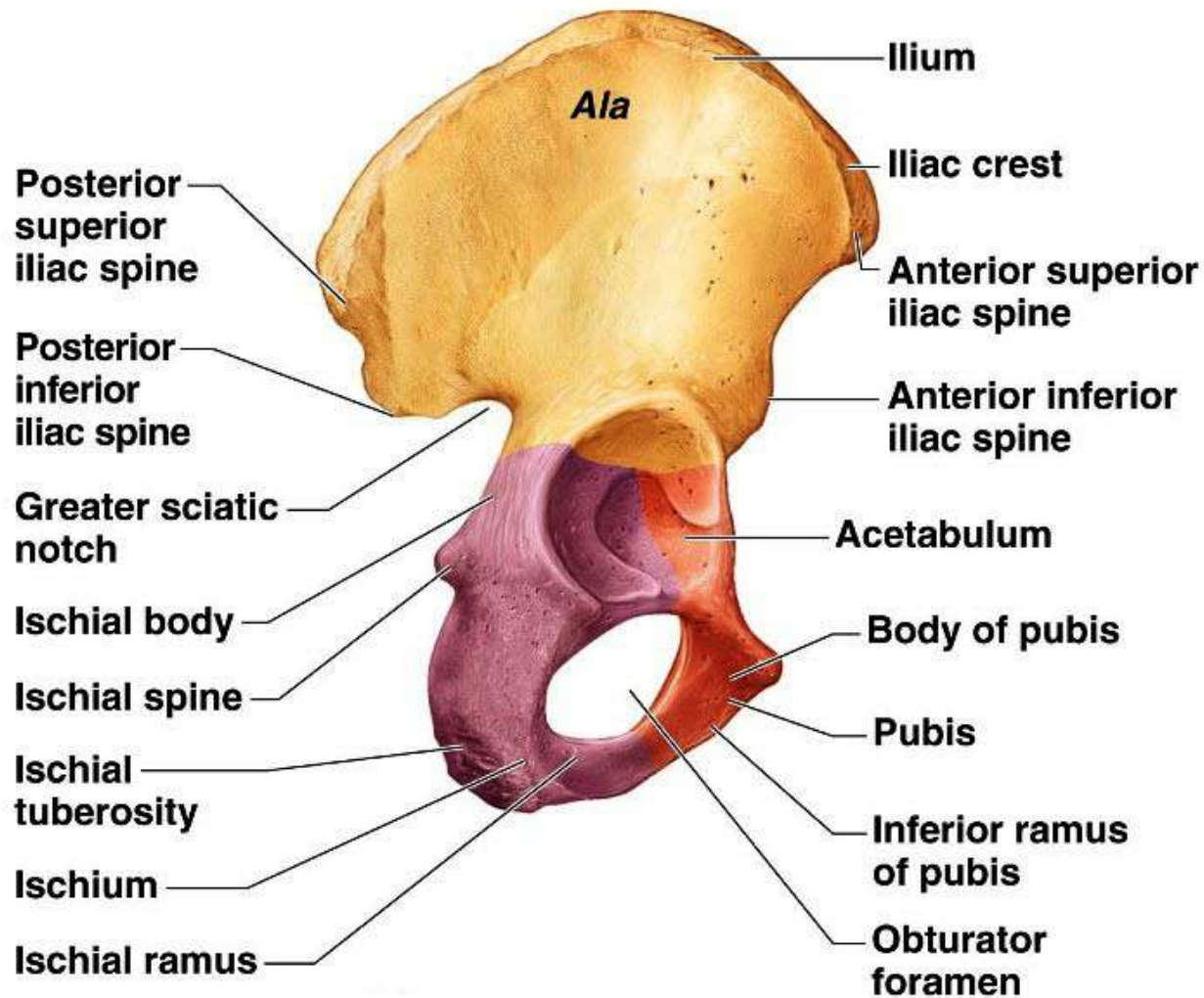
The Pelvis

The hip bones articulate **posteriorly**, with the sacrum at the **sacroiliac joint**; they also meet with one another **anteriorly**, at the **symphysis pubis**.

Ant.



Post.



ilium:

Is the upper flattened part of the bone, possesses:

- ❖ iliac crest ; ends in front at the **ant.sup.iliac spine** & ends behind at **Post. sup. iliac spine**
- ❖ iliac tubercle ; 2 inches behind Ant. Sup.iliac spine.
- ❖ Ant.inf.iliac spine ; below the ant. sup. Iliac spine.
- ❖ post.inf.iliac spine ; below the post.sup.iliac spine
- ❖ Gluteal lines; on the outer surface of ilium (ant, post., inferior)
- ❖ Greater Sciatic notch; behind the acetabulum.
- ❖ iliac fossa: inner surface of ilium.

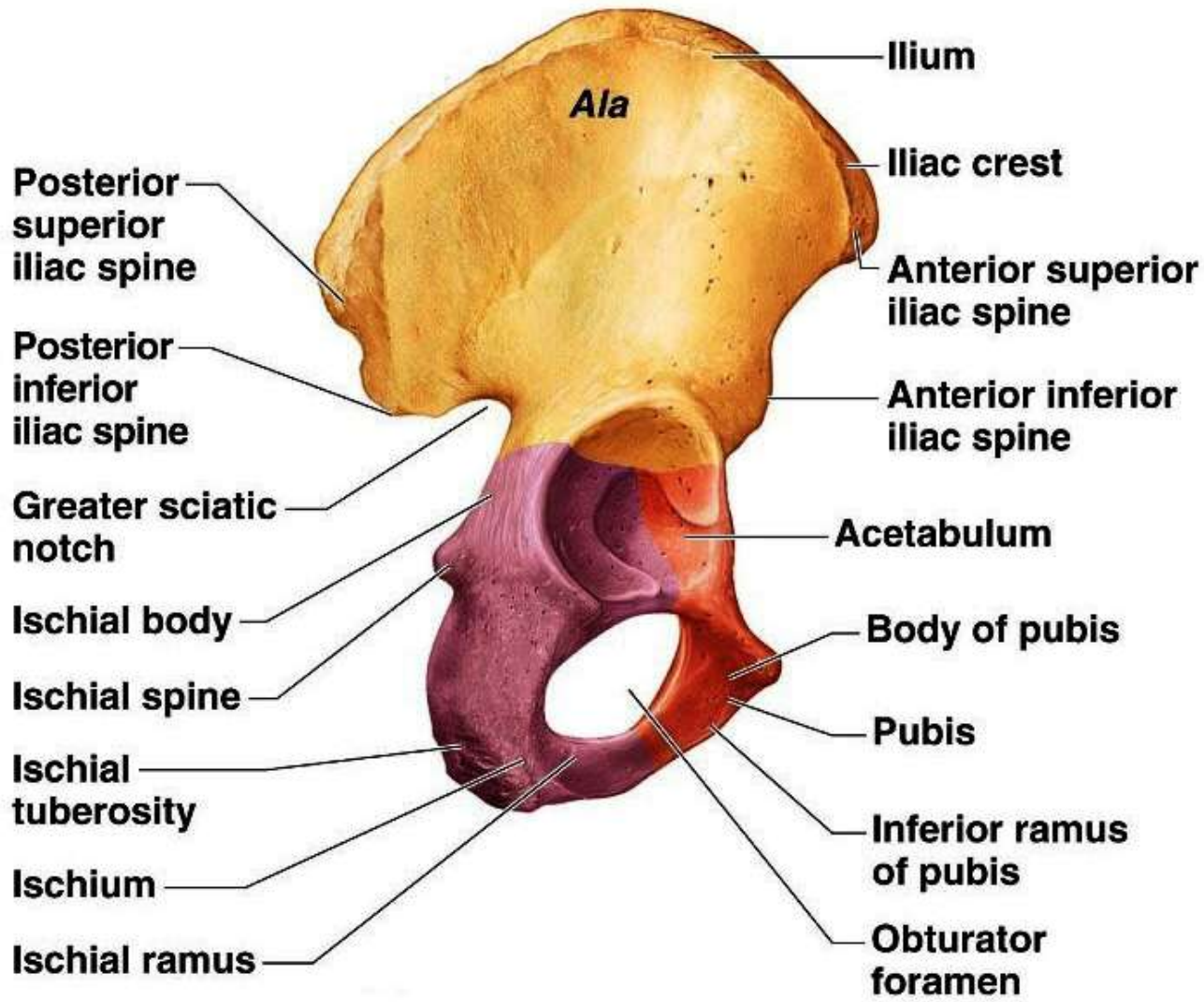
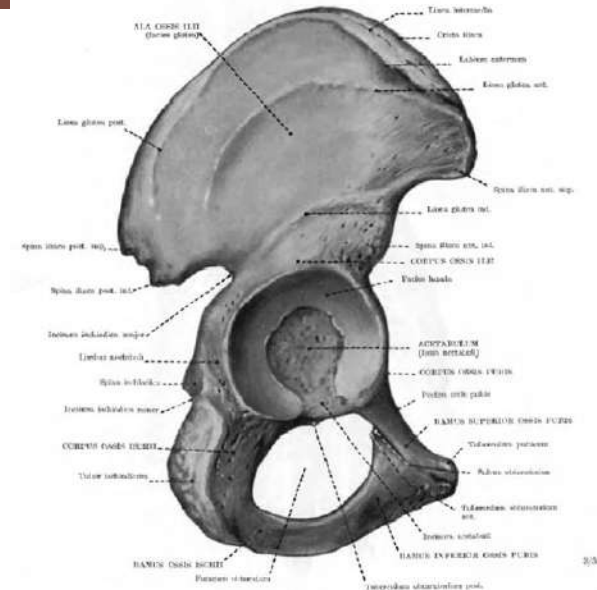
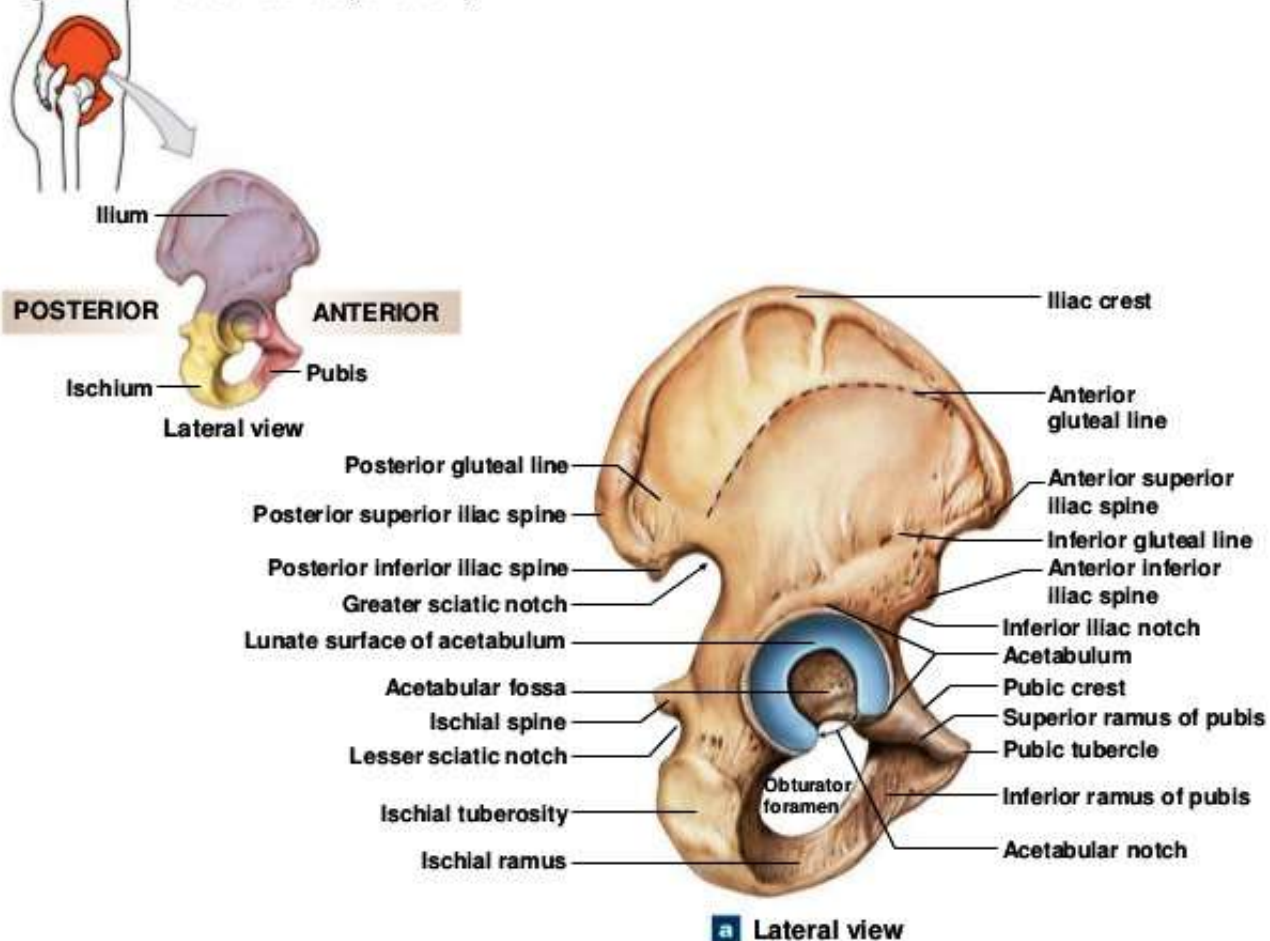
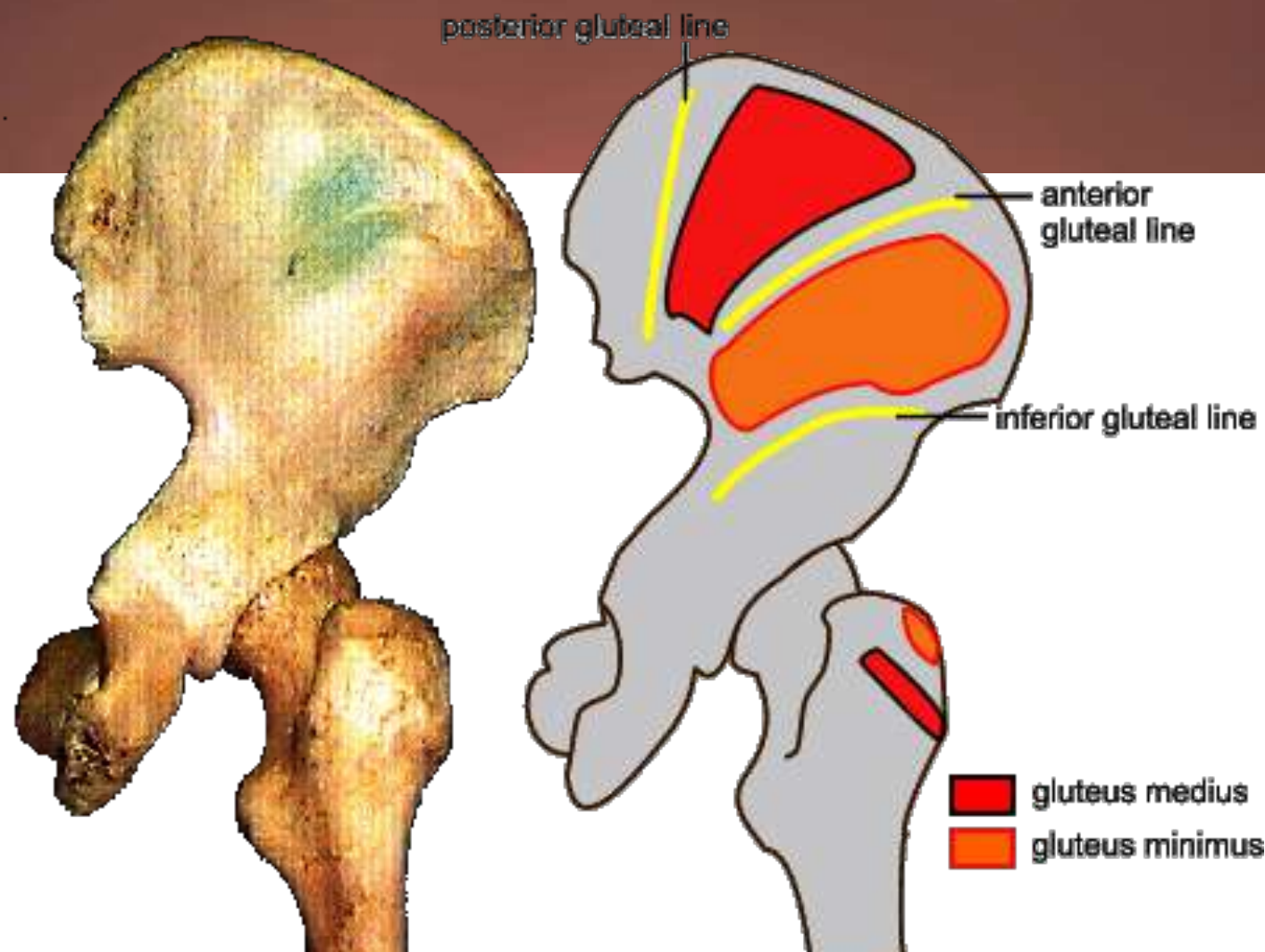


Figure 7.10a The Pelvic Girdle (Part 1 of 2)

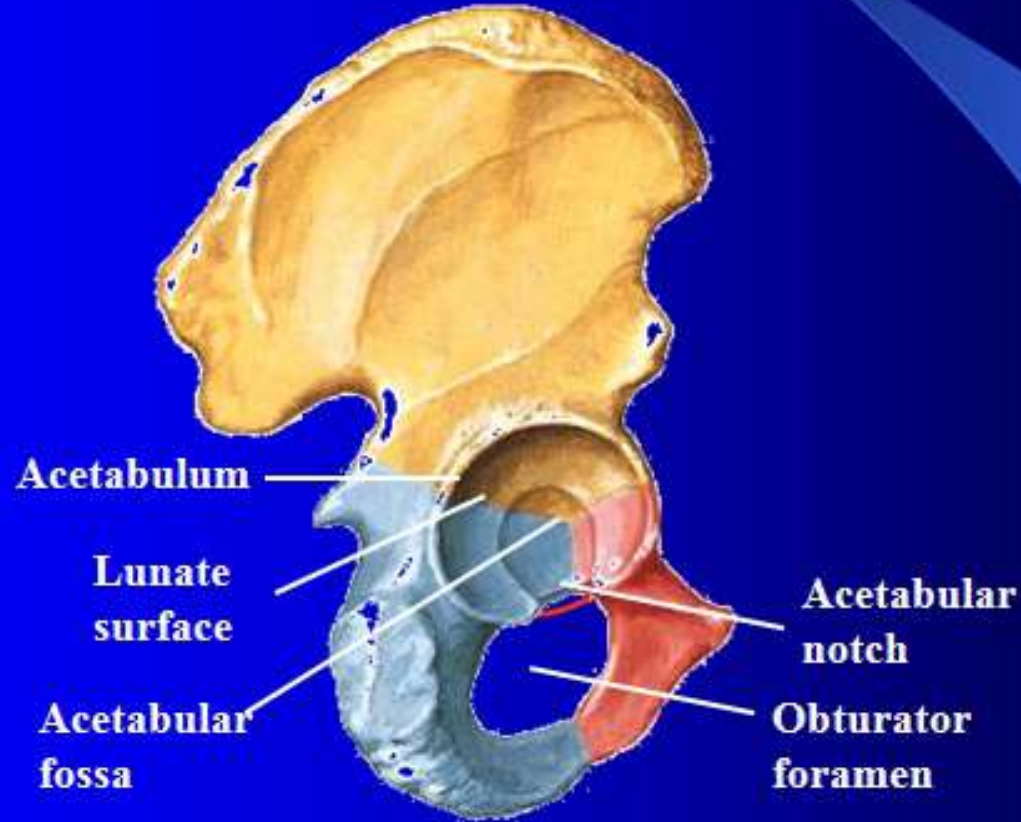


Gluteal lines: inferior, anterior(middle) posterior lines on the outer surface of ilium.



Acetabulum:

- Located on the outer surface of hip bone.
- It articulates with head of the femur to form hip joint.
- **Acetabular notch**; deficient inferior margin.
- **Acetabular fossa**: floor of acetabulum, non articulating.
- **Horse-shoe-shaped articulating surface**, covered with hyaline cartilage.



Pubis:

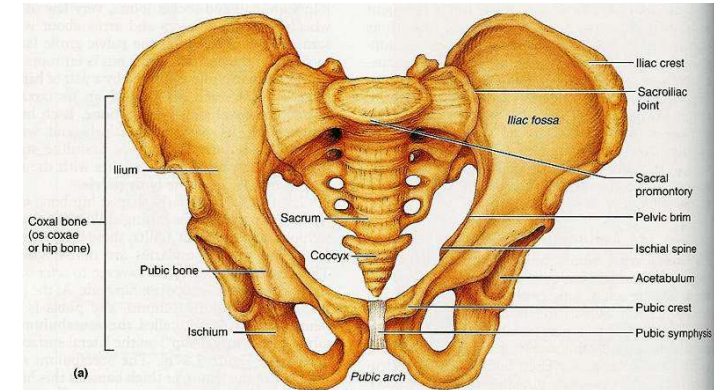
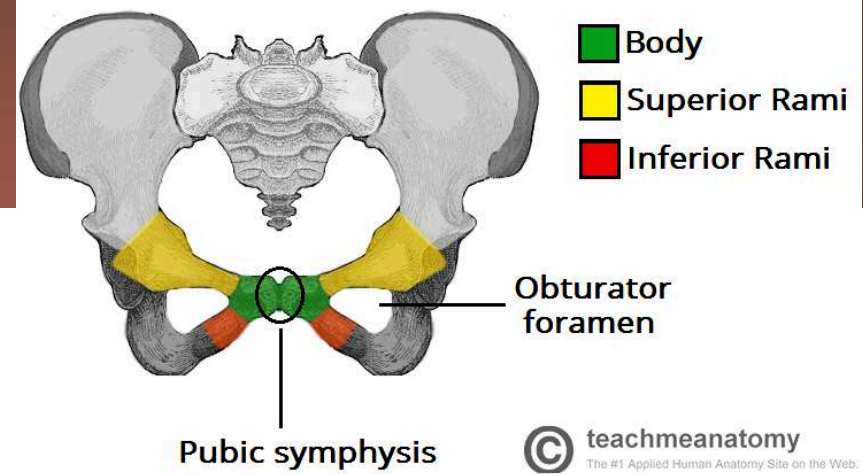
Divided into:

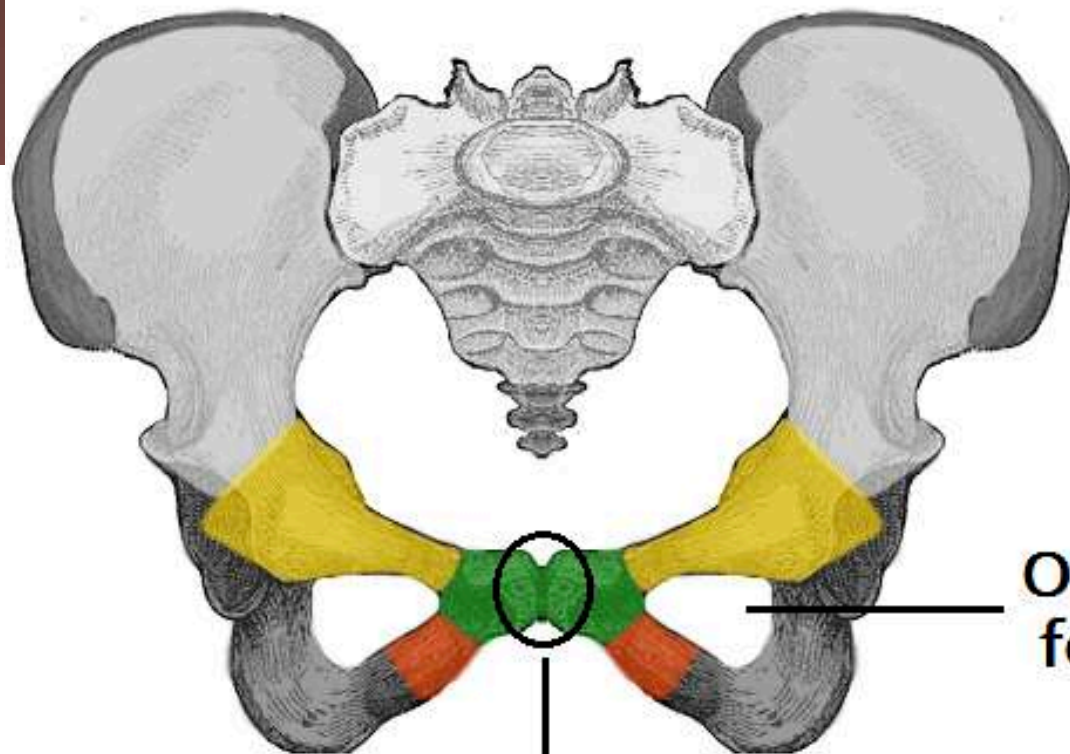
1-body

2-Superior ramus

3-Inferior ramus

- The bodies of the two pubic bones articulate with each other in the midline anteriorly at the **symphysis pubis**.
- The sup. Ramus joins the ilium & ischium at **acetabulum**.
- The inferior ramus joins the ischeal ramus below the **obturator foramen**.
- **Pubic crest** (the upper border of the body of the pubis) , it ends laterally at **pubic tubercle**. مهم





-  Body
-  Superior Rami
-  Inferior Rami

Obturator
foramen

Pubic symphysis



teachmeanatomy

The #1 Applied Human Anatomy Site on the Web.

Ischium

Is L-shape.

Consists of :

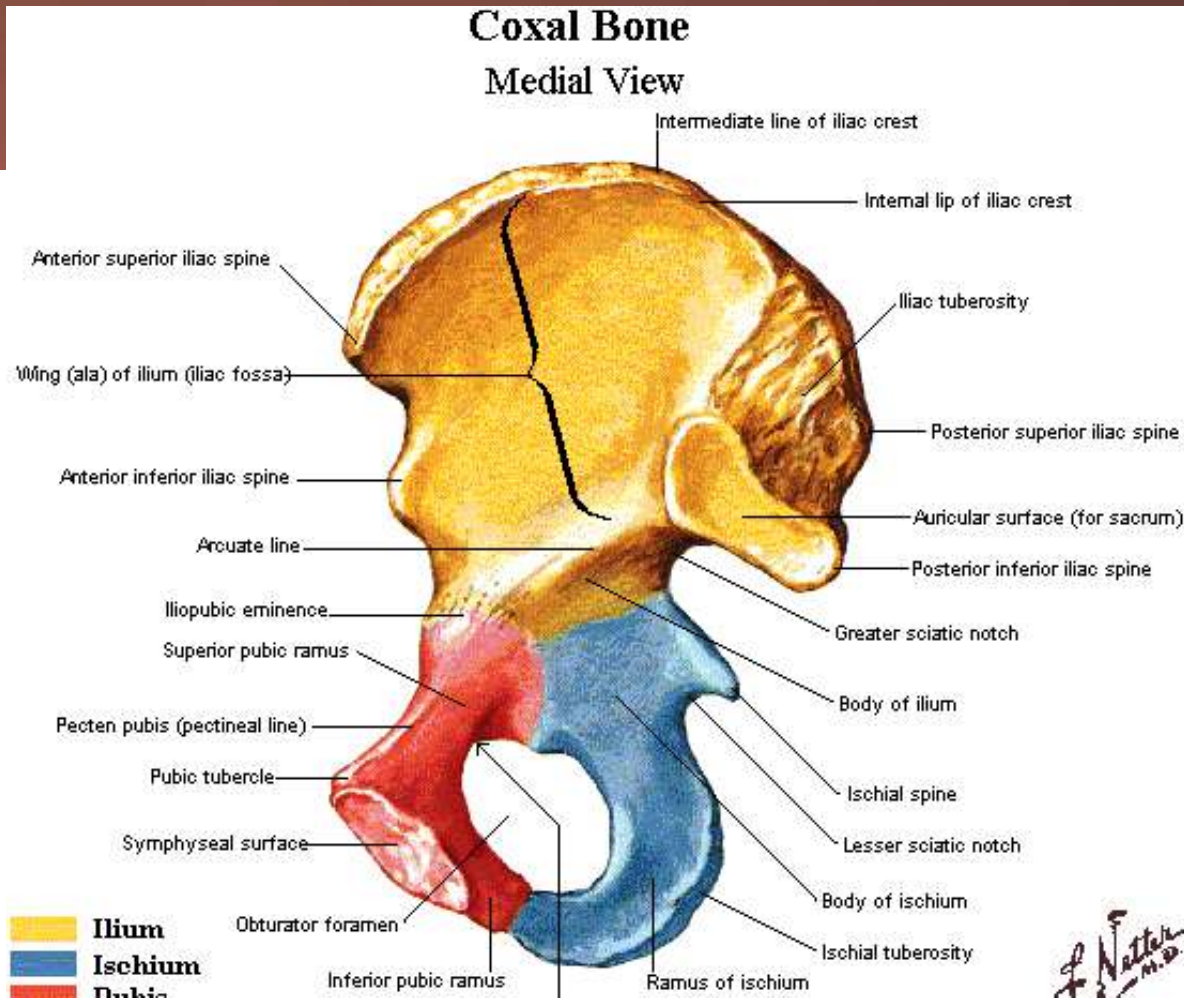
1- Body ; thick

2- Ramus; thin

❖ Ischeal spine

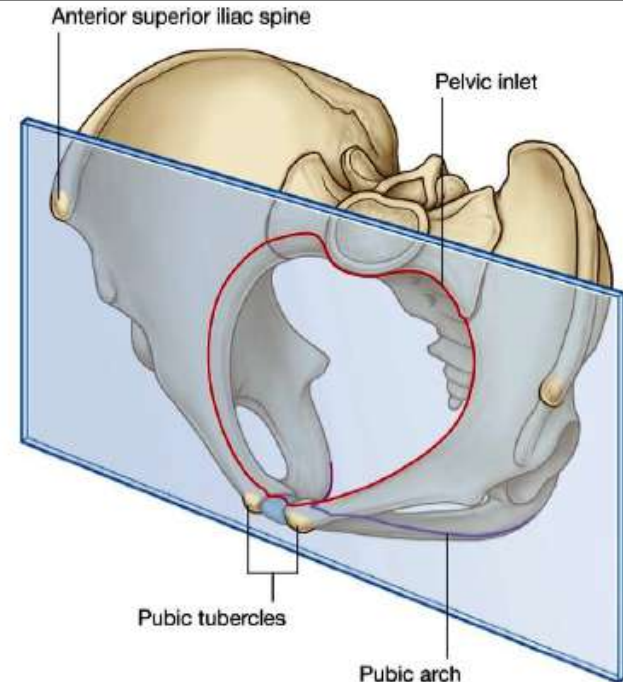
❖ Ischeal tuberosity

(post. Aspect of
lower part of body)



Anatomical position of pelvis:

In anatomical position, front of symphysis pubis & ant. sup. iliac spine lie in the same vertical plane.



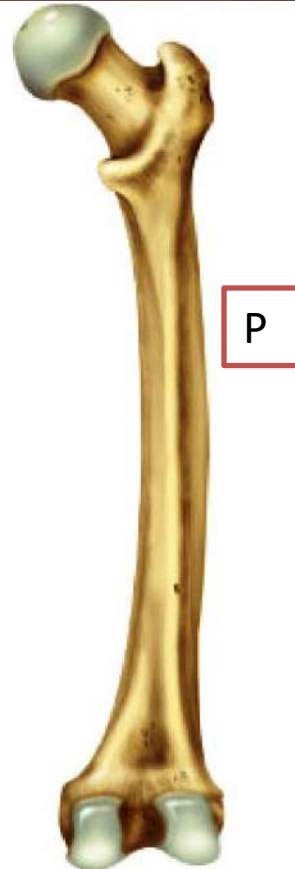
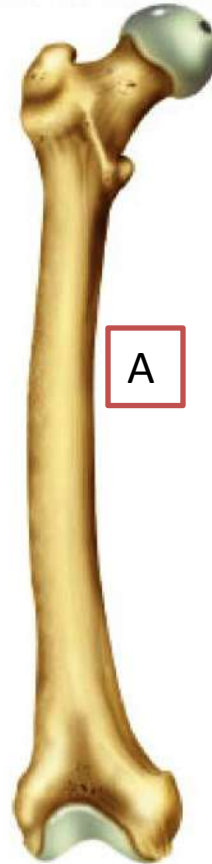
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Figure 5.26 Orientation of the pelvis (anatomical position).

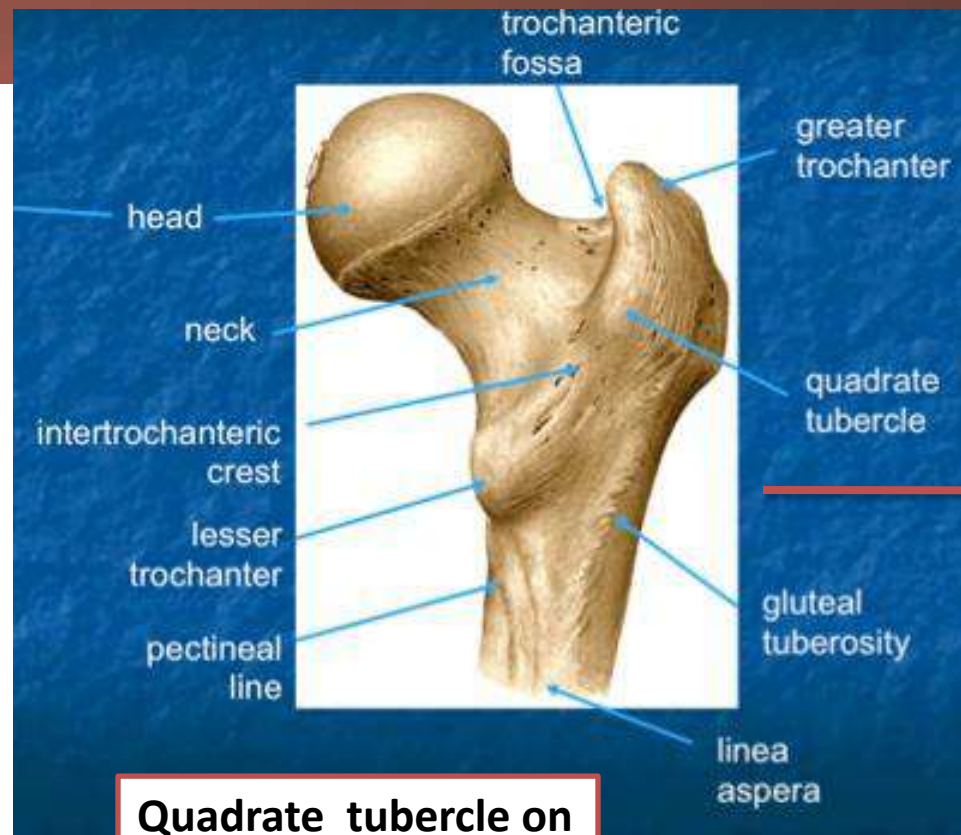
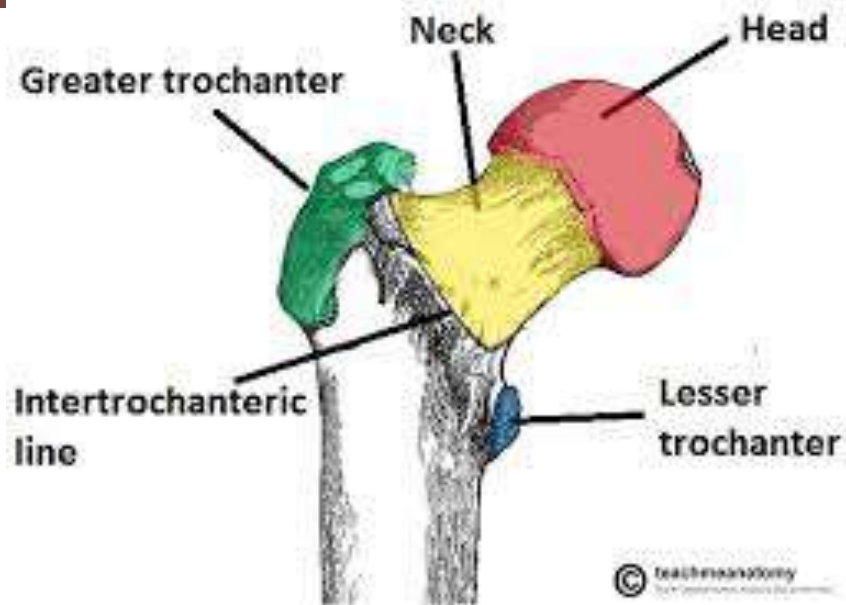
2-Femur

❑ It articulates **above** with acetabulum, to form the hip joint and **below** with the tibia & the patella to form the knee joint.

❑ The upper end:

- head
- neck; connects head to shaft
- greater & lesser trochanters
- Intertrochanteric line; connecting the two trochanters anteriorly.
- Intertrochanteric crest: connecting the two trochanters posteriorly.





Quadratus tubercle on the intertrochanteric crest posteriorly.

A**B**

□ The Shaft:

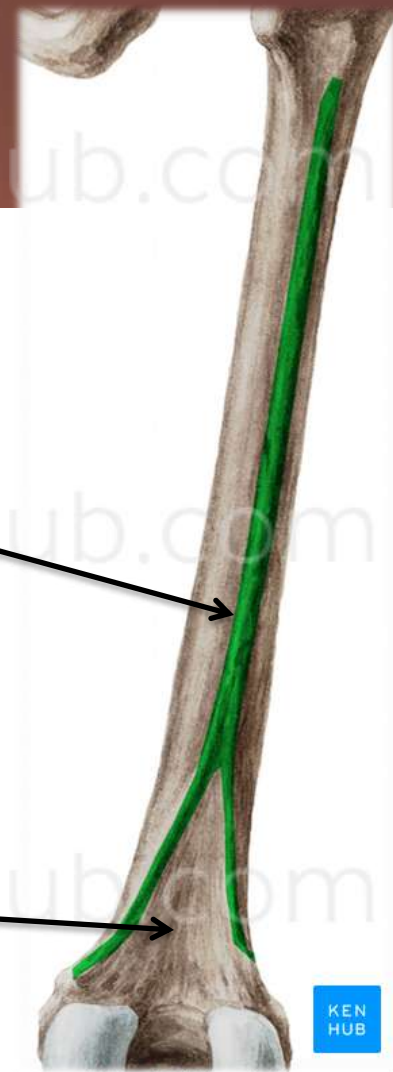
a. Anterior surface of the shaft:

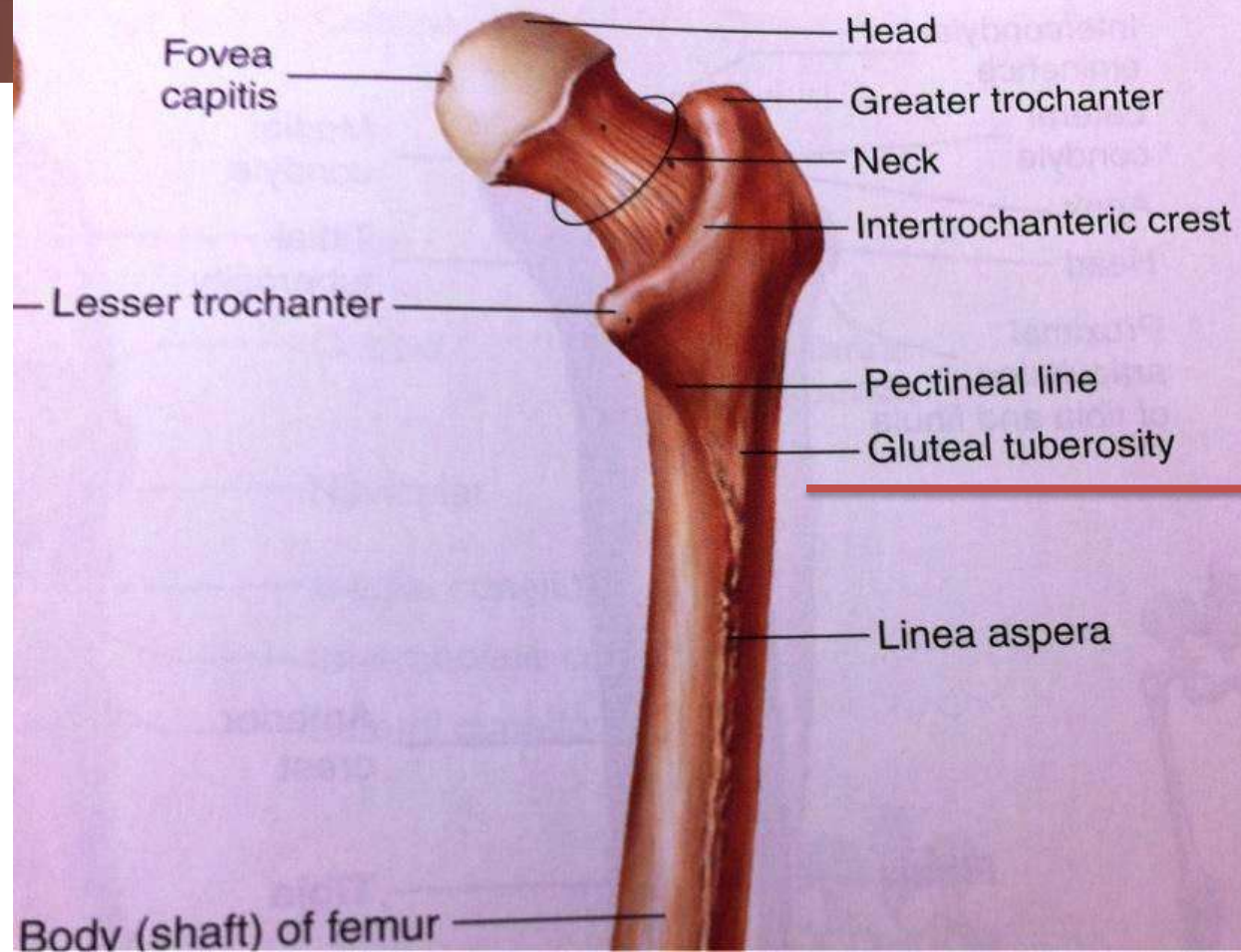
-Smooth & rounded.



b. Posterior surface of the shaft

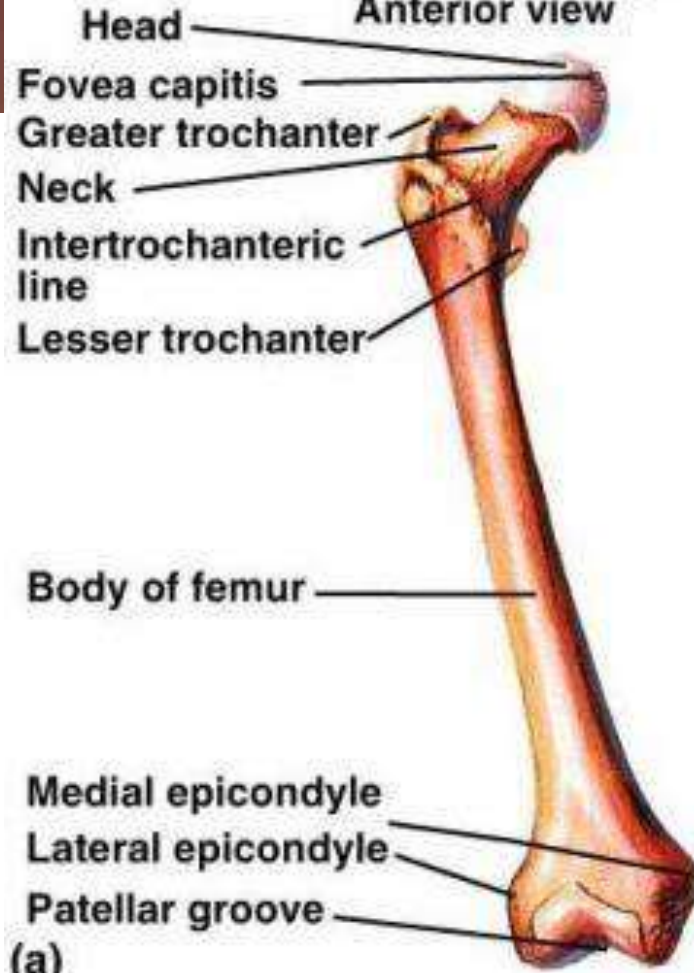
- ❖ Below greater trochanter is the gluteal tuberosity.
- ❖ has a ridge; linea aspera, which diverge above & below. Its medial margin continues below as the **medial supracondylar ridge**, to the **adductor tubercle** on the medial condyle. Its lateral margin continues below as the **Lateral supracondylar ridge**.
- ❖ Toward the distal end; the shaft becomes broader to form popliteal surface.



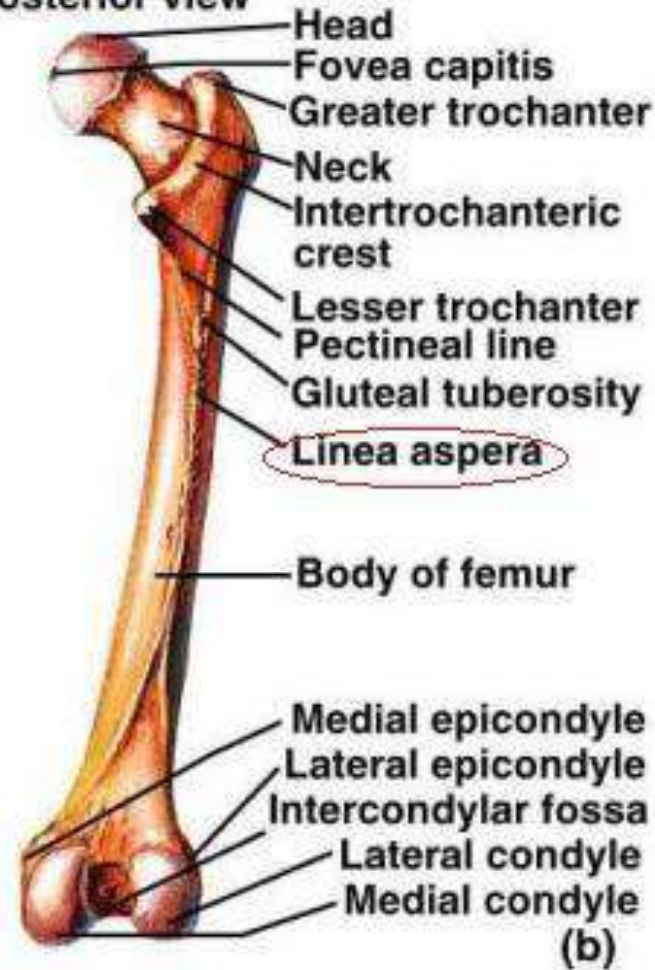


Right Femur

Anterior view

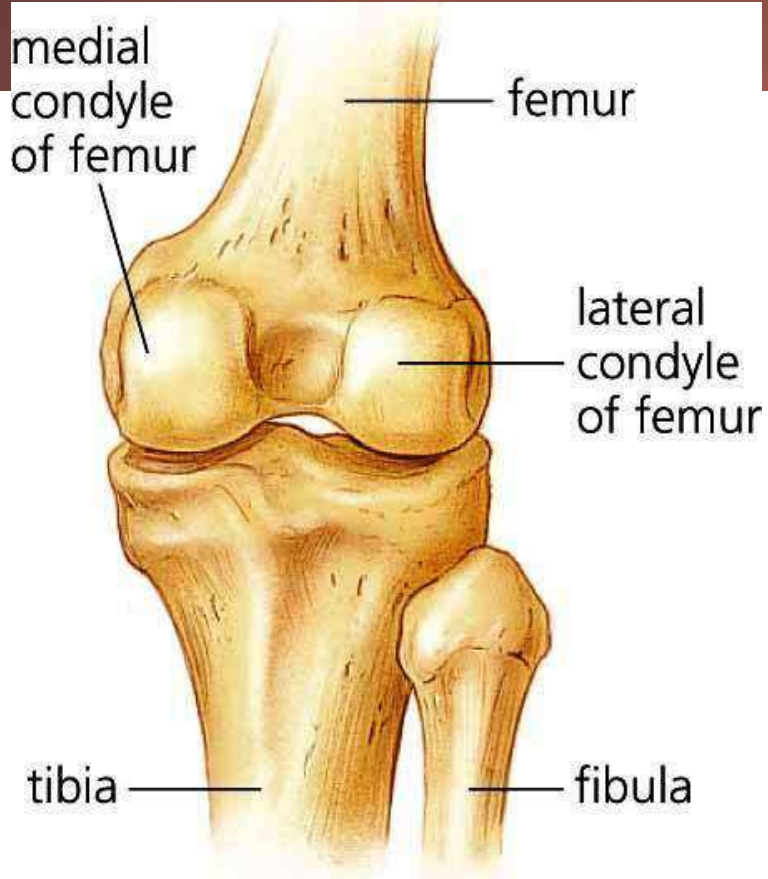


Posterior view

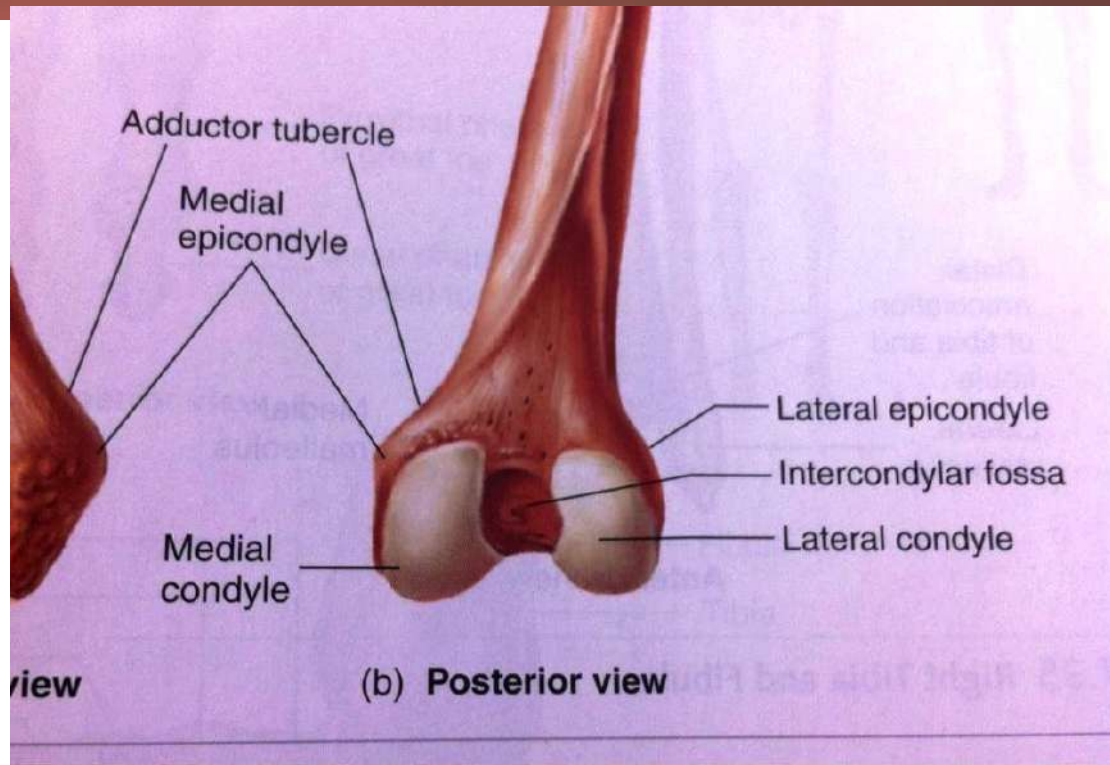


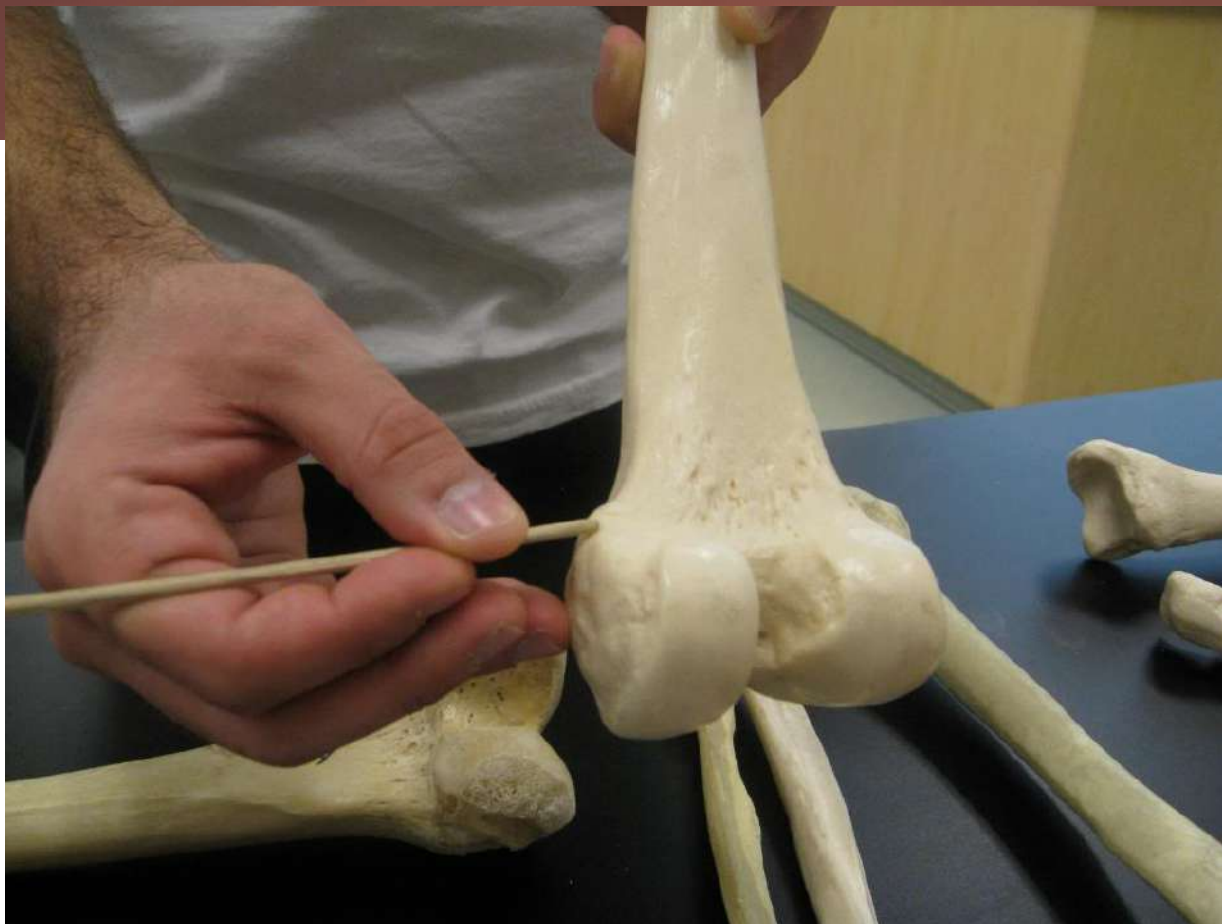
❑ Lower end

- ❖ Has lateral (larger) and medial condyles.
- **Anteriorly:** Their anterior surfaces are joined by an articular surface for the patella (***patellar surface***).
- **Posteriorly:** The two condyles are separated by the intercondylar notch.
- ❖ Above the condyles are medial & lateral epicondyles.
- ❖ Adductor tubercle is continues with the medial epicondyle.



Carlyn Iverson





Adductor
tubercle

Bones of the leg

☐ Patella

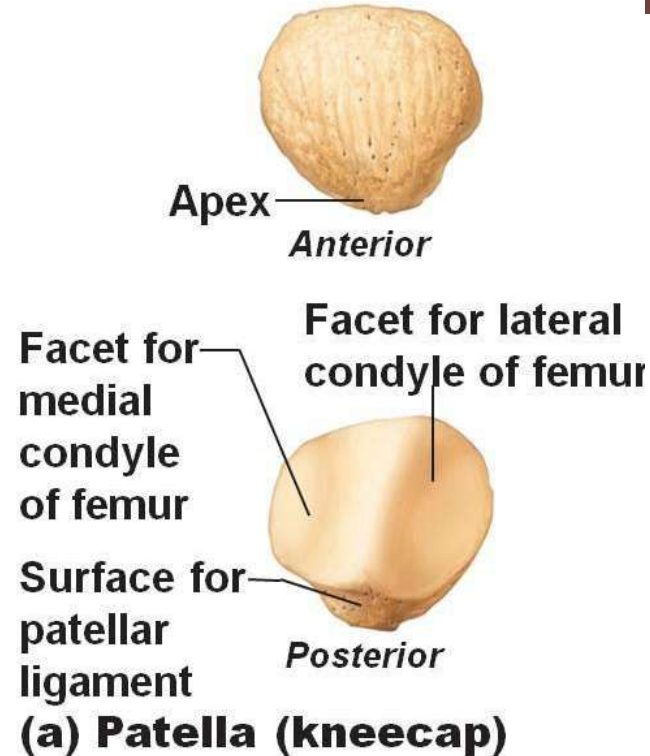
☐ Tibia

☐ Fibula

☐ Foot skeleton

Patella

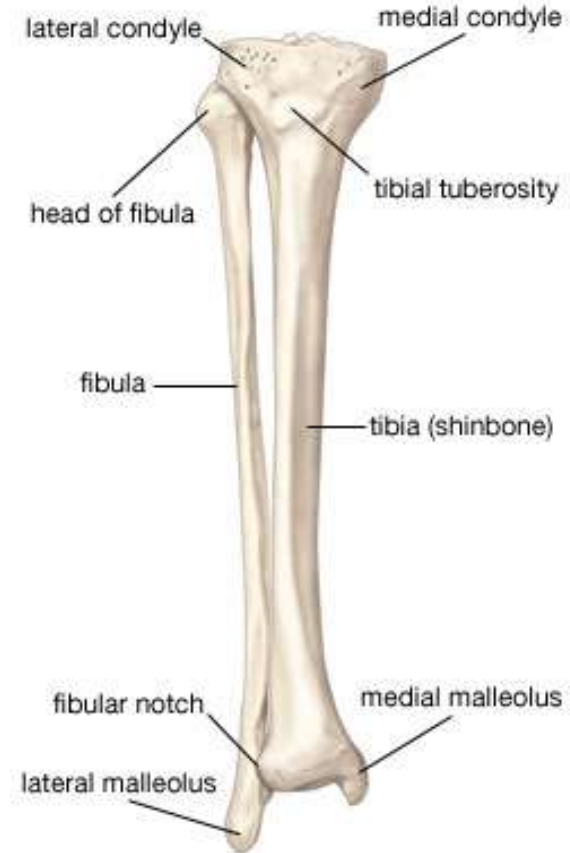
- Is the largest Sesamoid bone (develops with the tendon of **quadriceps femoris muscle** in front of the knee joint).
- It is triangular in shape.
- **Patella has:**
 - Apex(inferiorly)
 - Base (superiorly)
 - (2) surfaces
 - Ant:** rough longitudinal striae, separated from skin by subcutaneous bursa.
 - Post:** smooth, articular area separated by vertical ridge; facet for lateral(deeper/broader) and medial condyles of femur.
 - (3) borders: **sup, lateral& medial.**





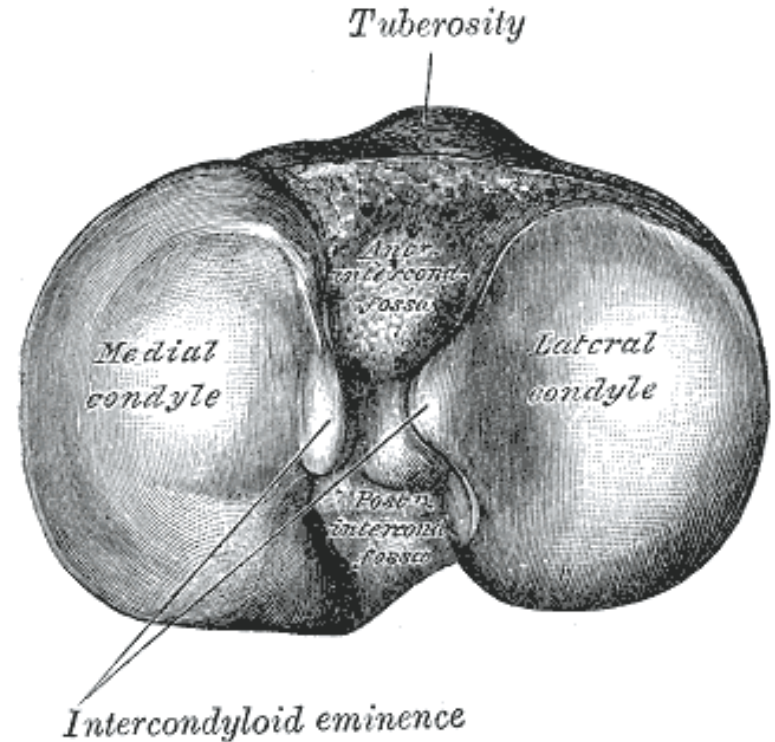
Tibia

- Is the large, weight- bearing **medial bone** of the leg.
- It articulates with the Condyles of the femur & the head of the fibula above and with the talus & the distal end of the fibula below.
- It has;
 - 1-Expanded upper end.
 - 2-Smaller lower end.
 - 3-Shaft.

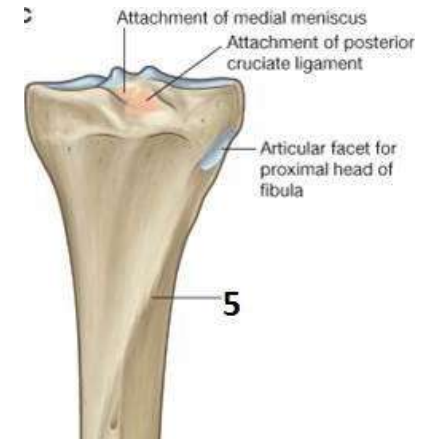
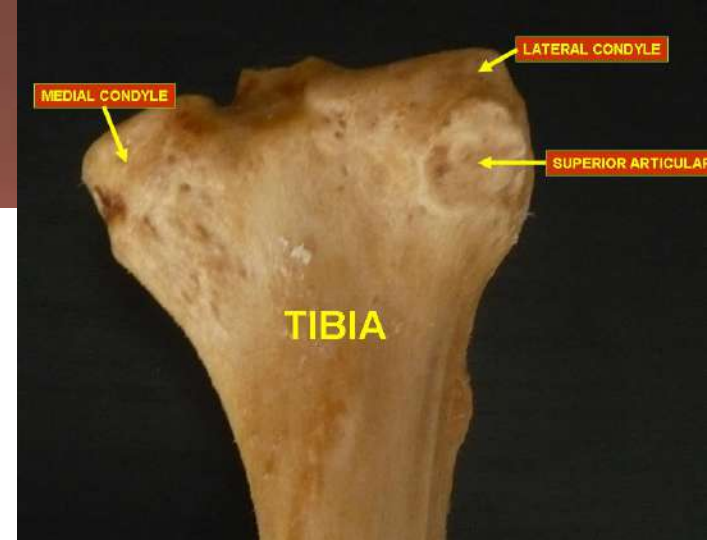


1-Upper end of tibia:

- **Lateral & medial (larger) condyles**, articulate with lateral & medial condyles of femur (Lat. & medial menisci intervening).
- **Anterior & posterior intercondylar areas.**
- **Intercondylar eminence.**



- Small circular articular facet for the head of the fibula on the lateral aspect of the lateral condyle.



2-Shaft of tibia:

➤ Is triangular in cross section.

➤ 3 borders

(ant., medial, interosseous(lateral))

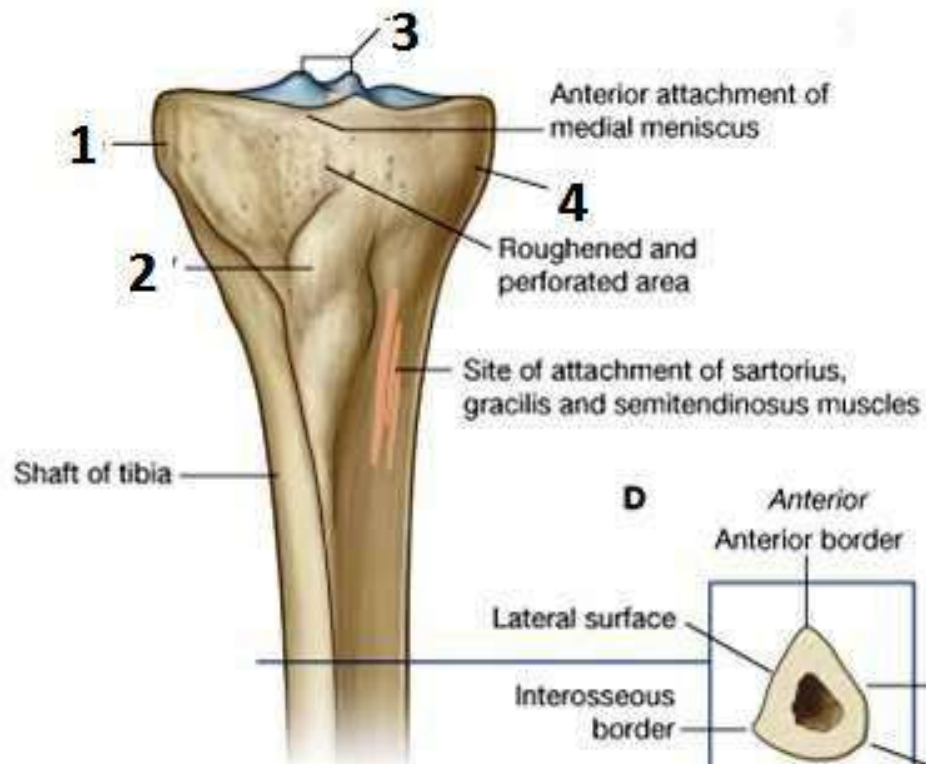
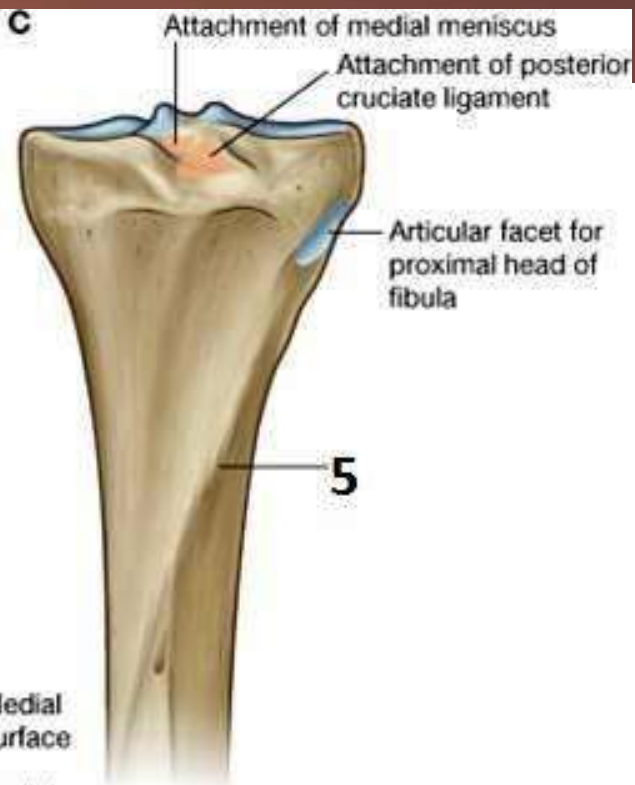
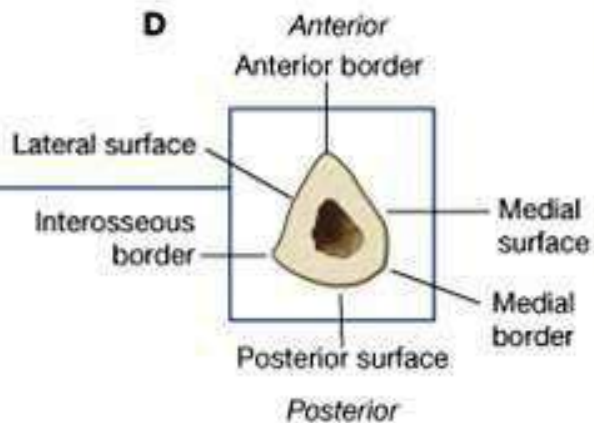
➤ 3 surfaces

(medial, lateral, posterior)

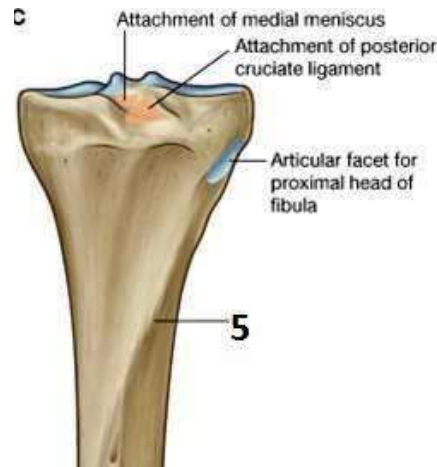
➤ Ant. & medial borders with medial surface between them are subcutaneous.

➤ Ant. Border is prominent & forms the shin.

Ant.border become rounded below, it becomes continuous with medial malleolus.

B**C****D**

- **Tibial tuberosity**; on the *anterior* border, receives attachment of ligamentum patellae.
- **Soleal line**; oblique line, on the upper **posterior** surface of shaft of tibia, for **Soleus** muscle attachment.



3-Lower end :

- Slightly expanded & projects medially downwards as the medial malleolus.
- **has 5 surfaces.**
- Wide rough depression on its **lateral surface** for articulation with fibula.
- Inferior surface shows saddle-shaped articular surface for the talus.



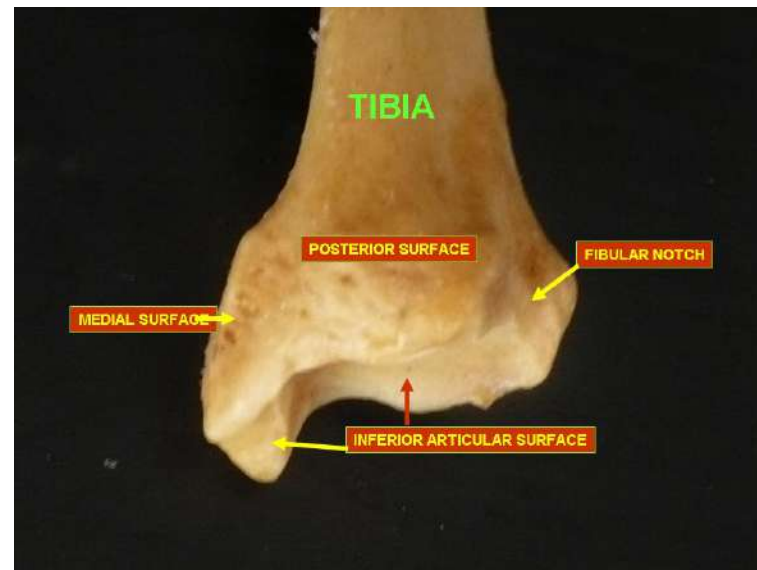
ANTERIOR



POSTERIOR

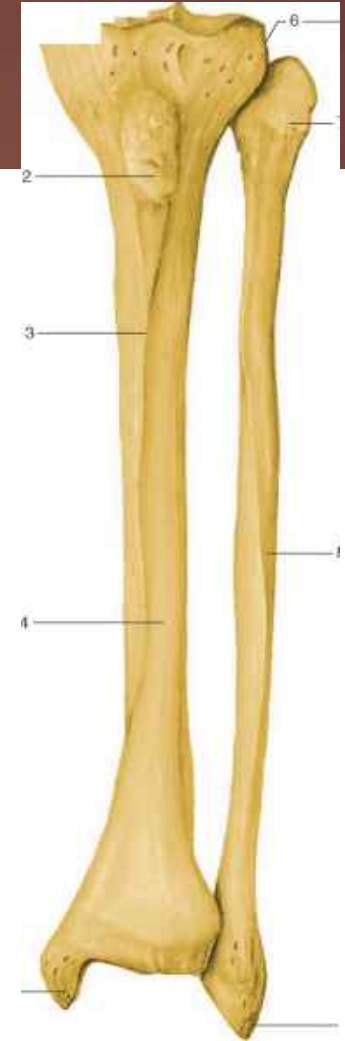


LATERAL



Fibula

- ❖ Is the slender **lateral bone** of the leg.
- ❖ It takes no part in articulation at knee joint.
- ❖ It forms the lateral malleolus of ankle joint.
- ❖ It takes no part in the transmission of body weight.
- ❖ It provides attachment for the muscles.
- ❖ It has upper end, shaft and lower end.



A. Upper End:

☐ **Head;**

- Slightly expanded
- has articular surface for articulation with lateral condyle of tibia.
- Neck

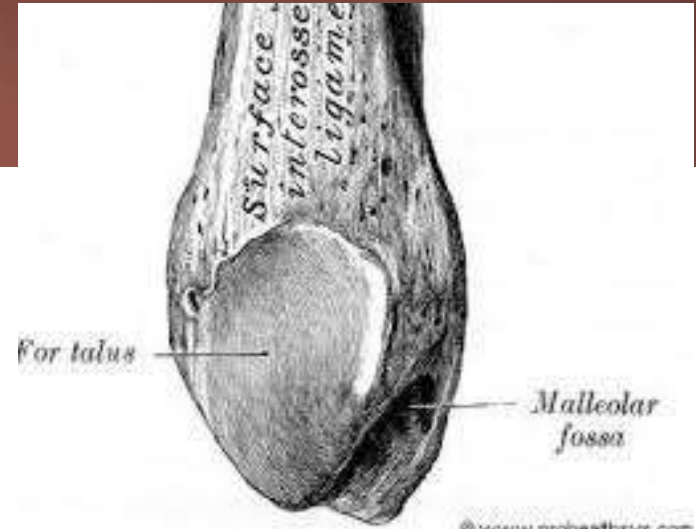
☐ **Styloid process (apex of the head);** blunt elevation projects upwards from its post.surface.

B-Shaft:

- Is long & slender, has 4 borders & 4 surfaces.
- medial (interosseous) border, gives attachment to interosseous membrane.
- Anterior border is continuous below with the lateral malleolus.

C- Lower end:

- Forms the triangular **lateral malleolus**, which is **subcutaneous**.
- on the medial surface for lateral malleolus is a **triangular articular facet** to articulate with the lateral surface of talus.
- **Malleolar fossa.**
- articular facet for articulation with the lower end of tibia forming inferior tibio-fibular joint.



Fibula

Upper end:

- fibular head
- neck of fibula

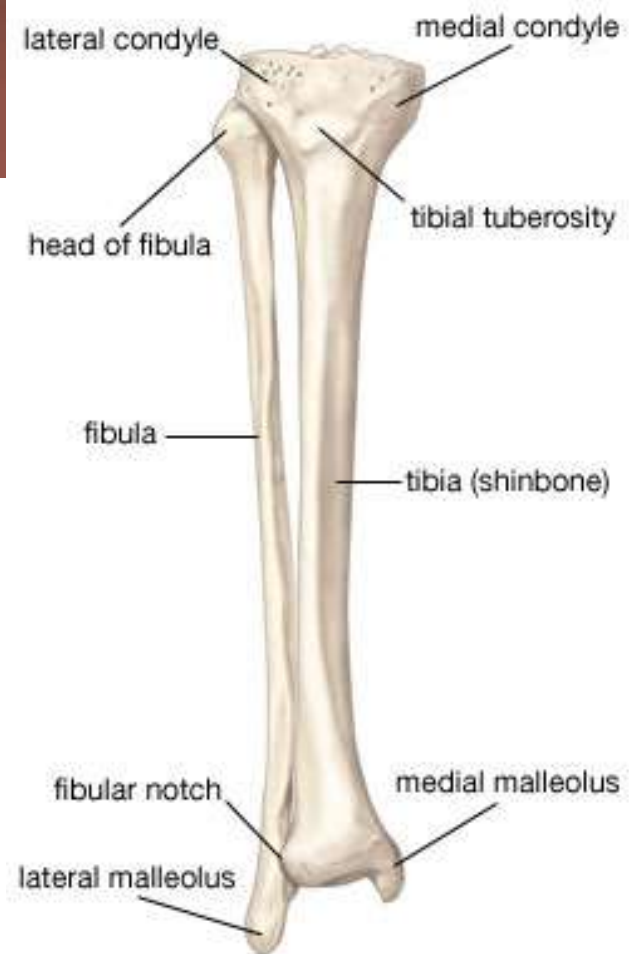
Shaft:

- interosseous border

Lower end:

- lateral malleolus





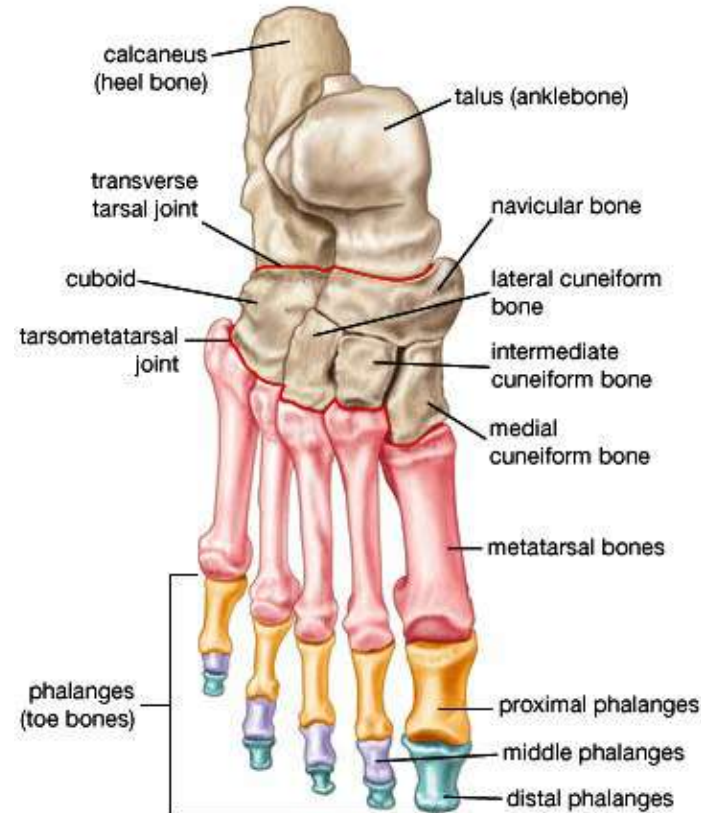
Skeleton of the foot

The bones of the foot is divided into:

1-Tarsal bones

2- Metatarsal bones

3-Phalanges



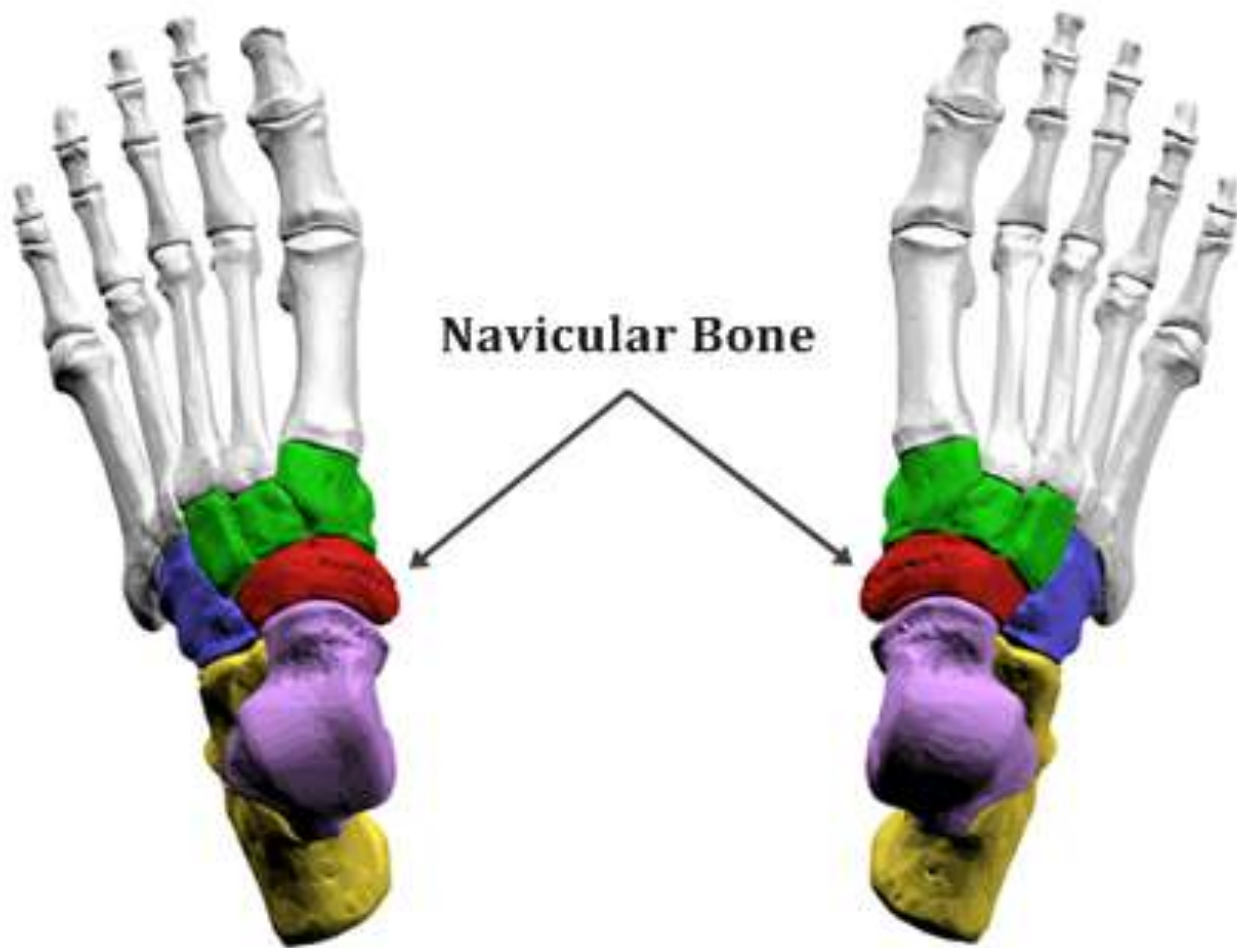
1-Tarsal bones:

Are formed of 7 bones:

a-proximal(2)

b- intermediate(1)

C-distal (4)



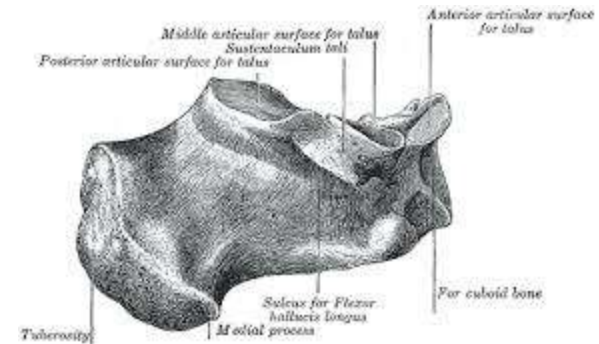
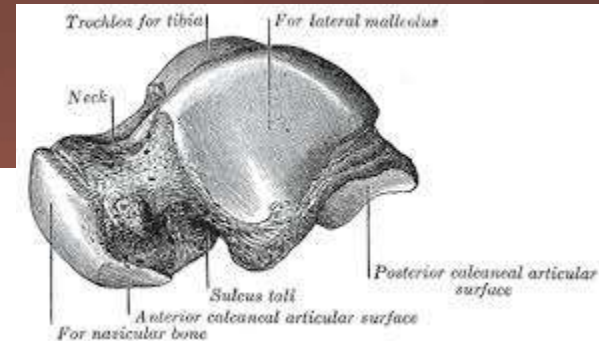
Proximal tarsal bones (2)

1-Talus

- articulate above at the ankle joint with tibia and fibula and in front with **navicular bone**.
- possess head, neck& body.

2-Calcaneum

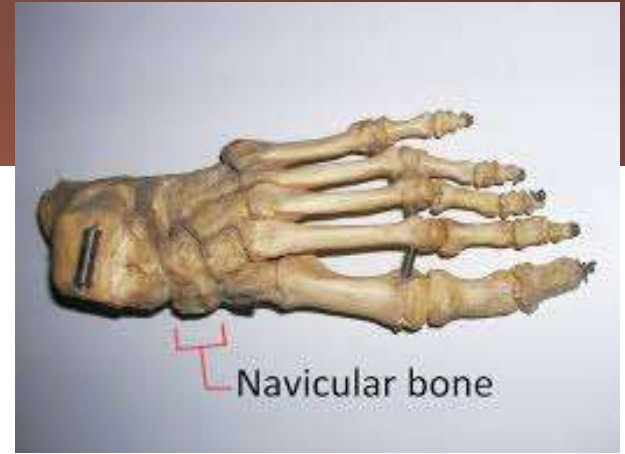
- largest bone in the foot.
- forms the prominence of the heel.
- articulates above with **talus**& in front with **cuboid**.
- has 6 surfaces



Intermediate tarsal bone(1)

Navicular bone

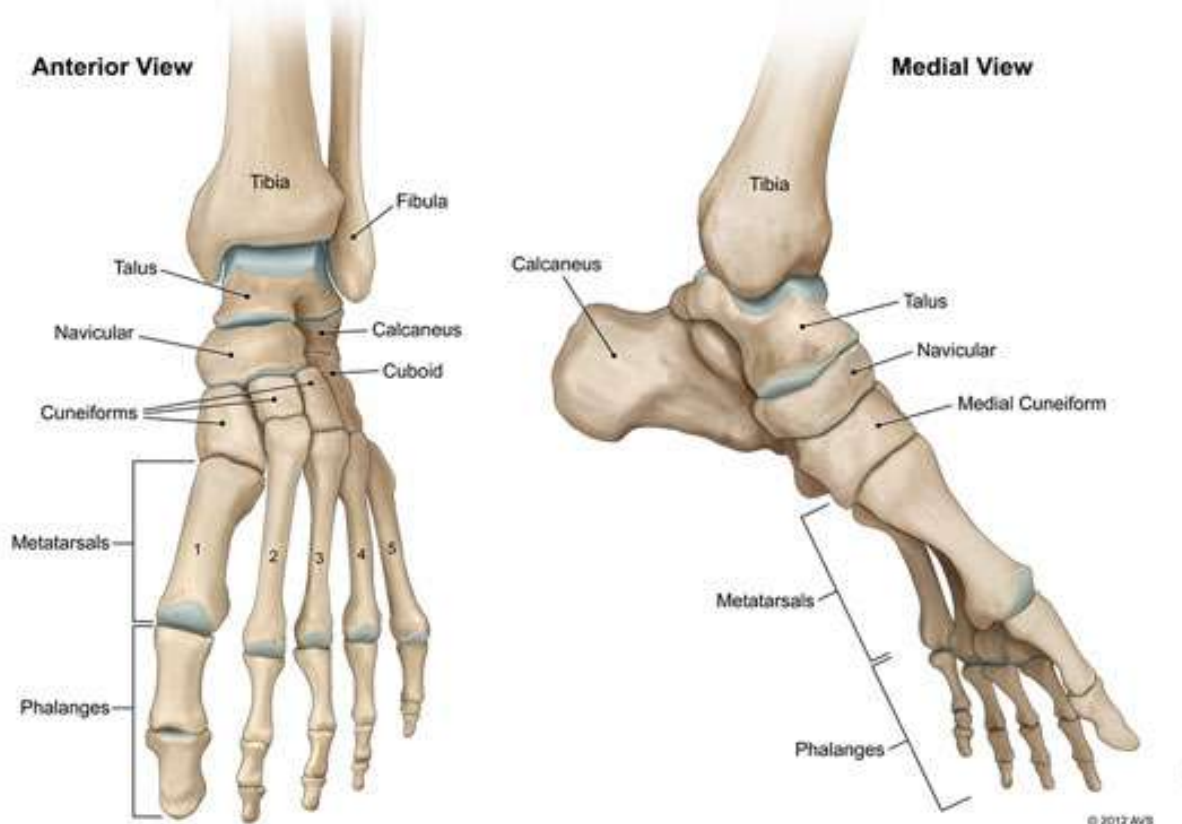
Located between the head of talus & cuneiform bones.
-has prominent projection called **navicular tuberosity**.

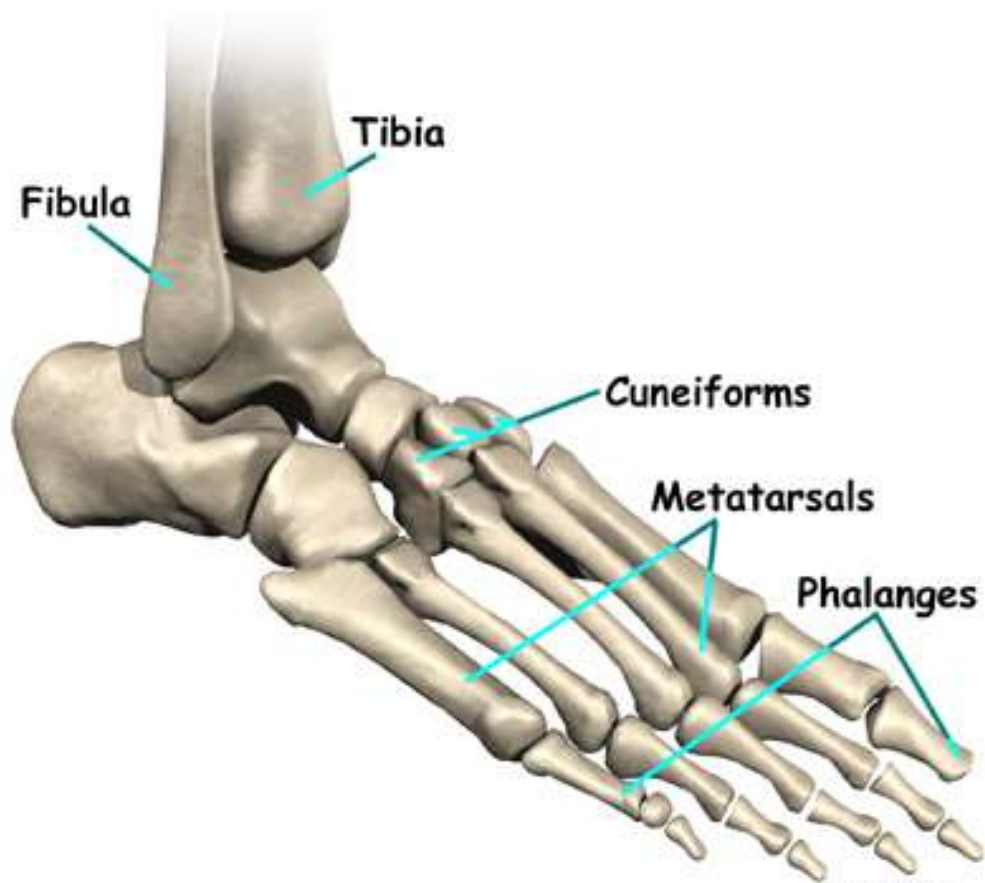


- Distal tarsal bones
- Cuniforms (3)
 - These are 3 wedge-shaped bones, articulate proximally with Navicular bone & distally with the first 3 metatarsal bones.
 - Medial(largest),intermediate (smallest) & lateral cuniforms.**



Normal Left Foot and Ankle





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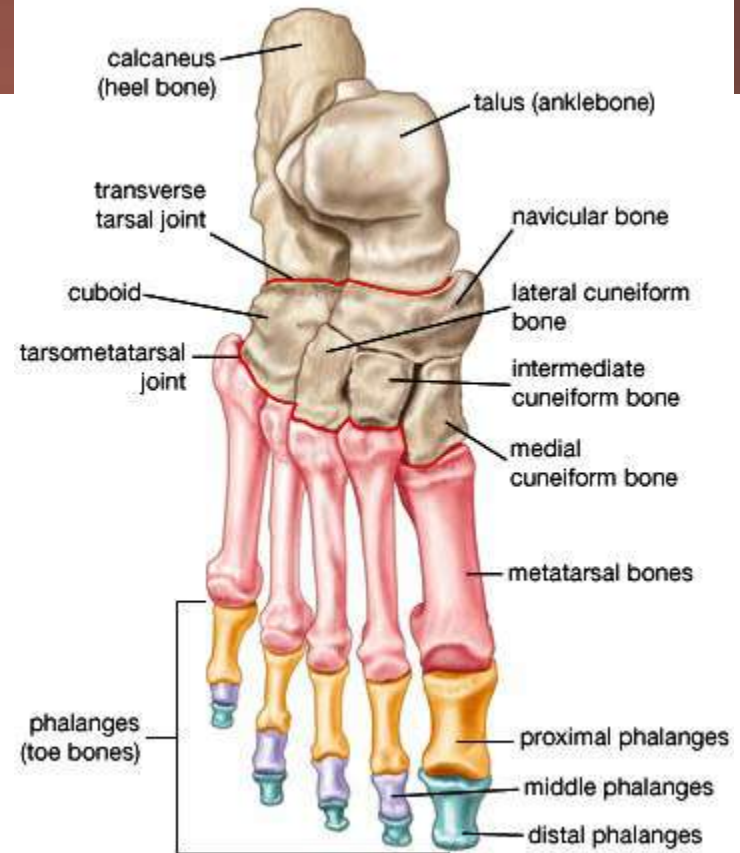


- **Cuboid (1)**

- **Is the most lateral in distal row.**

- Situated between calcaneum & 4th, 5th metatarsal bones

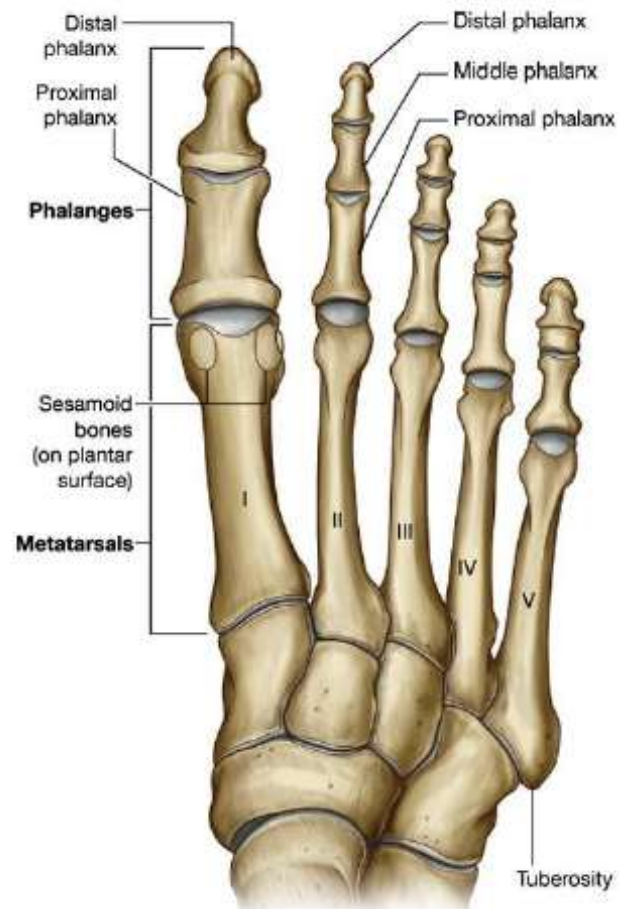
- tuberosity



2-Metatarsals

- 5 in number
- Connecting tarsus to phalanges.
- Numbered from **medial side**.
- They have; base, shaft and head.

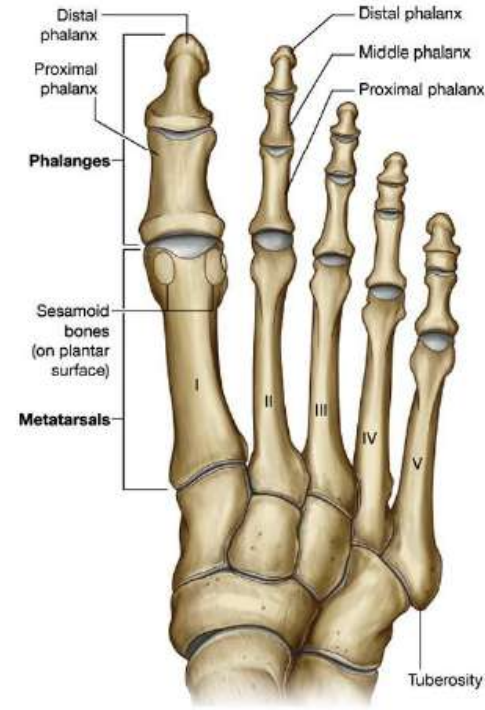




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Figure 6.95 Metatarsals and phalanges. Dorsal view.

3-Phalanges

- Each toe has three **phalanges** EXCEPT the big toe, which has only two (proximal and distal)
- Each phalanx has **head** (distally), **shaft** and a **base** (proximally).
- The five phalanges just like the metatarsals are numbered from medial to lateral side (start numbering from the big toe)



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Figure 6.95 Metatarsals and phalanges. Dorsal view.

Arches of the foot:

3 arches for foot

A- (1) Transvers arch

B- (2) longitudinal arches (medial / lateral).

Importance?

Two important functions for foot;

1- support body weight 2- serve as a lever for propulsive forward body movement in walking & running.

-foot is not a single bone but formed of a series of small bones, connected by joints.

-A segmented structure can hold up weight only if it is built in the form of an arch.

-Medial longitudinal arch is the largest and the most important clinically.

-Shape of bones, strong ligaments, tone of muscles, are important factors in arch support.

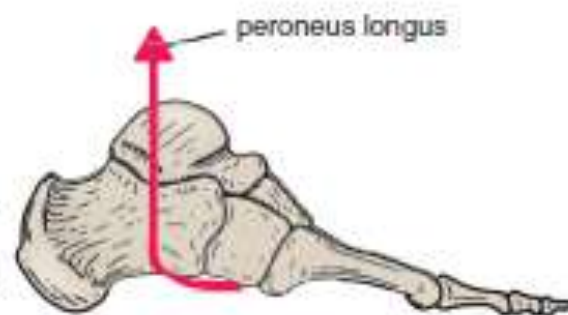
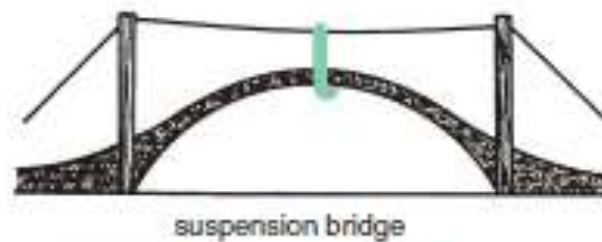
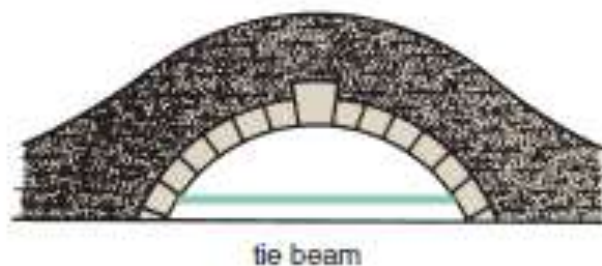


FIGURE 10.68 Different methods by which the arches of the foot may be supported.

Transverse arch consists of bases of metatarsals bones, cuboidal and 3 cuniforms.

Lateral longitudinal arch consists of ; calcaneum, cuboid & 4th & 5th metatarsals bones.

Medial longitudinal arch consists of ; calcaneum, talus, navicular, three cuniforms bones & the first 3 metatarsals.

talus; the concave proximal surface of the navicular bone

The cuboid is the keystone.

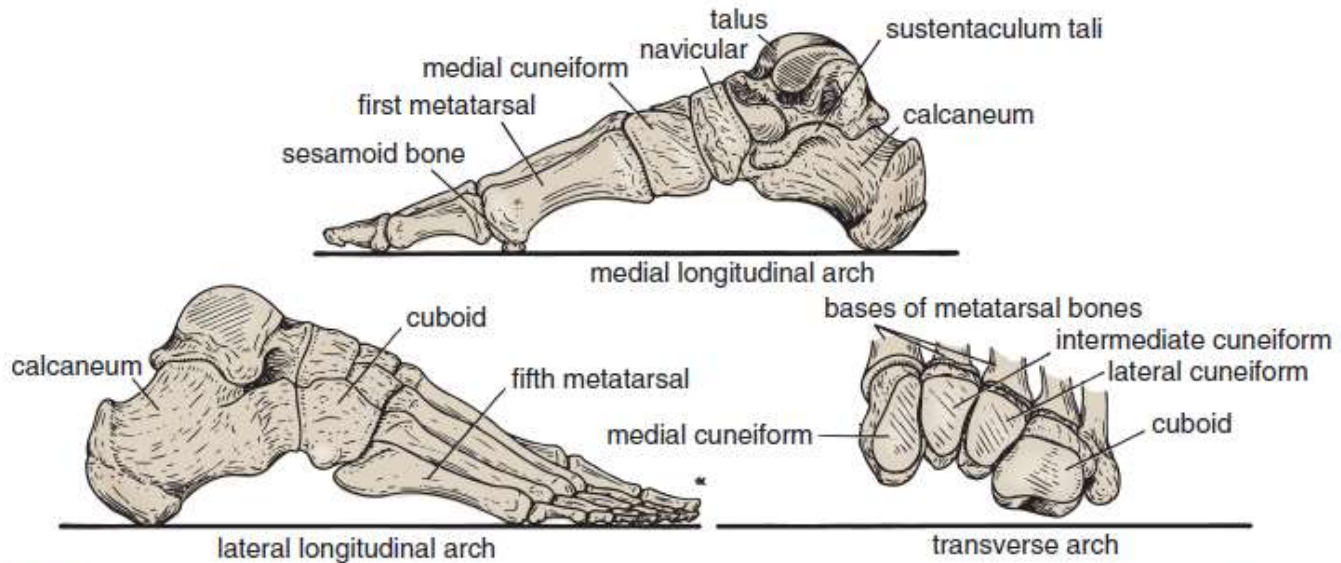
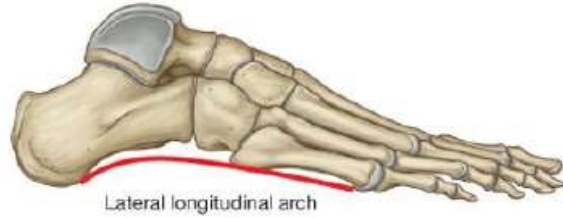
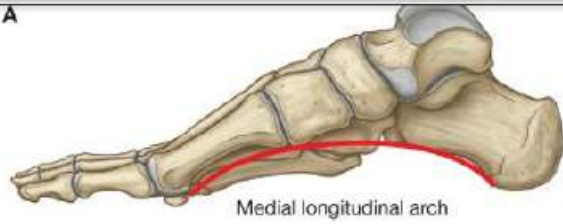


FIGURE 10.67 Bones forming the medial longitudinal, lateral longitudinal, and transverse arches of the right foot.

A**B**

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Figure 6.107 Arches of the foot. **A.** Longitudinal arches, right foot. **B.** Transverse arch, left foot.

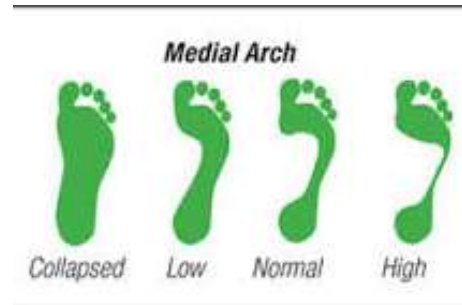
Clinical problems associated with arches of foot.

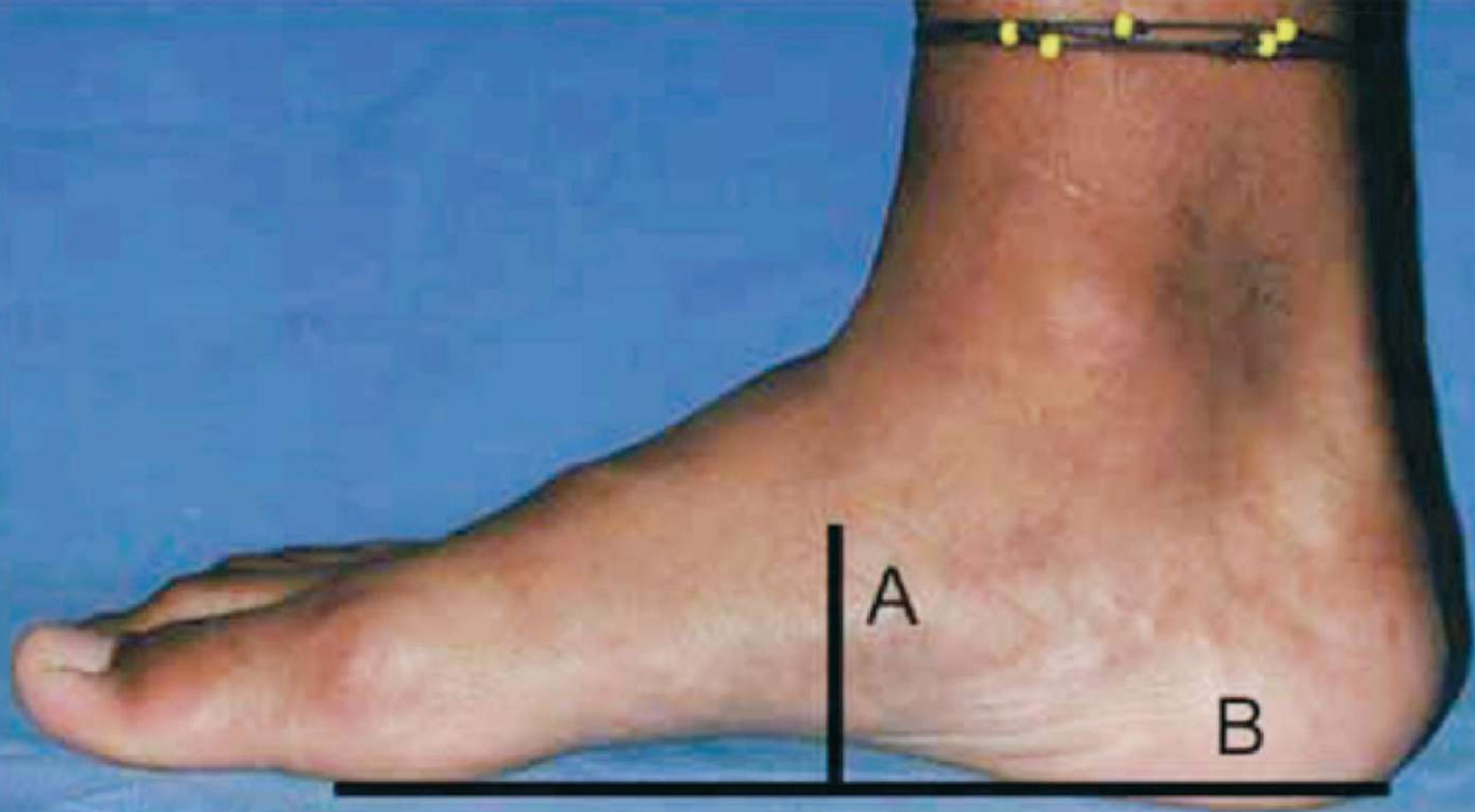


Excessive exercise, standing for long periods, overweight and diseases,,,causes weakness of muscles and stretch of ligaments which produce pain.

Pes planus (flat foot): is a condition in which medial longitudinal arch is depressed.

Pes Cavus (clow foot) is a condition in which medial longitudinal arch is unduly high.







Pes planus



Pes Cavu

Thank you!

